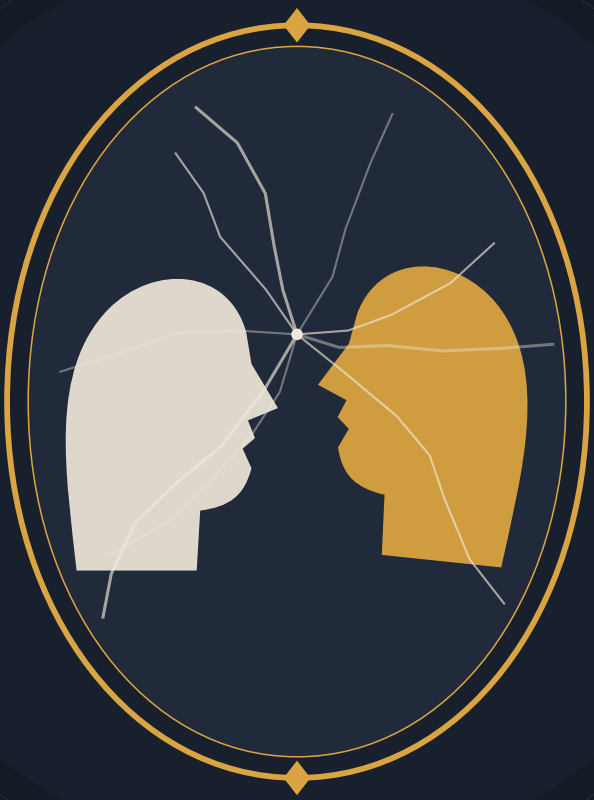
THE PSYCHOLOGY OF SELF-DECEPTION

The Mind That *Lies* to Itself



Why smart people defend beliefs against the evidence — and how to catch your own mind in the act

REAL CASE STUDIES · REFLECTION EXERCISES · EVIDENCE-BASED

The Mind That Lies to It-self

Why intelligent people defend beliefs against the evidence, and how to catch your own mind in the act

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The Mind That Lies to Itself

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A note on content. This book is intended for general education. It is not a substitute for professional psychological, medical, legal or financial advice. Case studies are drawn from published research and news reporting, cited in the Notes. Everyday scenes involving named individuals such as “Priya” or “Dr Suresh” are composites, created to illustrate common patterns, and do not describe real people.

A note on evidence. Psychological research is revised as new evidence emerges. Findings that are disputed, culturally variable or often exaggerated are flagged in the boxes headed “What the research does not say.”

For everyone who has ever said "I was wrong"
and meant it.

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Introduction: The Liar Who Believes Himself

There is a particular kind of lie that never feels like lying.

You know the ordinary kind. A child with chocolate on his face says he hasn't touched the box. A colleague claims the email "must have gone to spam." These lies have two minds in them: the one that knows the truth and the one that tells the story. The teller feels the gap between them, and that gap is what makes lying uncomfortable.

This book is about the other kind, the lie with only one mind in it. The teller and the listener are the same person, and somewhere along the way the gap closes. The story stops feeling like a story. It starts feeling like the plain truth.

You have watched other people do this. The uncle who has forwarded the same miracle cure to the family WhatsApp group for six years, untroubled by the fact that he still takes his blood-pressure tablets. The friend who has left three jobs because of "toxic bosses" and has not yet noticed the one element common to all three offices. The politician whose principles flip the moment his party changes its position, and who seems sincerely unaware that anything has flipped at all.

It is easy, and rather enjoyable, to watch other people do this. The harder and more useful exercise is to notice that you do it too. Not occasionally. Not only when you are tired or emotional. Routinely,

competently, and often most of all about the things you care about most.

That is the uncomfortable claim at the centre of this book: **the mental machinery that lets intelligent people believe false things is not a malfunction found only in foolish people. It is standard equipment.** It is fitted in every model, including the one reading this sentence.

Why a mind would lie to itself

At first the idea sounds absurd. The whole point of having a brain, surely, is to build an accurate picture of the world. A gazelle that misjudges where the lion is gets eaten. A farmer who fools himself about the monsoon loses his crop. Accuracy pays.

But accuracy is not the only thing that pays. Humans are social animals, and for most of our history survival depended less on reading the landscape than on reading one another. We needed allies. We needed a good reputation. We needed to belong to a group that would share food in a bad season and stand beside us in a quarrel. We needed, in short, to be people others trusted, liked and respected.

Those social goals put pressure on what we believe. A belief is not just a map of reality; it is also a badge of membership, a source of self-respect, and a position we have publicly defended. Changing it can cost us standing with our group, our sense of being a competent person, and the dignity of having been right. So the mind, it seems, does something clever and occasionally dangerous. It continues to process evidence, but it does so with a thumb on the scale.

Psychologists call this **motivated reasoning**: reasoning that is steered, usually without our awareness, by what we want to be true

rather than only by what the evidence shows. The term is associated especially with the social psychologist Ziva Kunda, whose influential 1990 review argued that people do not simply believe whatever they like. They are constrained by the need to build a justification that would seem reasonable to a neutral observer. The trick is that they search for that justification in a biased way, examine unwelcome evidence harder than welcome evidence, and stop looking as soon as they find something that supports the conclusion they preferred from the start.

That is why self-deception rarely feels like self-deception. It feels like thinking. We do gather evidence. We do weigh it. We simply do so the way a lawyer prepares a case rather than the way a judge decides one.

The inner press secretary

A helpful image comes from the psychologist Jonathan Haidt and others who have compared part of the reasoning mind to a press secretary. A press secretary does not set government policy. The decisions are made elsewhere, often for reasons the press secretary never hears. The job is to stand in front of the cameras and explain why those decisions were wise, consistent and entirely justified.

Much of what we call our "reasons" works in a similar way. Intuitions, feelings, loyalties and habits often arrive first. Then the press secretary, who is articulate and quick on its feet, produces the explanation. Crucially, the press secretary believes its own briefings. It is not cynical. It is loyal.

You should not take this metaphor too literally. Reasoning is not always an after-the-fact excuse; people can and do reason their way to new conclusions, especially when they are curious, when the

stakes for their identity are low, or when they argue with others who disagree. One of the more hopeful findings you will meet later in this book is that human reasoning often works better in conversation than in isolation. But the press secretary is a good starting point because it captures something that surprises most people: the feeling of having reasons is not good evidence that those reasons caused the belief.

What this book is, and what it is not

Two excellent books already cover the broad territory. Daniel Kahneman's *Thinking, Fast and Slow* explains the shortcuts of intuitive judgement, and Jonathan Haidt's *The Righteous Mind* explores how moral intuitions shape politics and religion. If you have read them, you will meet a few familiar studies here. But this is not a catalogue of cognitive biases, and it is not a textbook.

It has a narrower focus: **self-deception, and the everyday situations in which reasoning quietly turns into a defence of who we are.** The questions it asks are practical:

- Why does admitting a mistake feel physically painful, and what does the mind do to avoid it?
- How do beliefs become tied to identity, so that an argument about facts feels like an attack on the self?
- Why do we notice the evidence that suits us and walk past the rest?
- Why are our memories, which feel like recordings, so often edited?
- Why does being clever sometimes make the problem worse rather than better?
- How do groups, parties and families build shared beliefs that no single member would have invented alone?

- Why do false stories travel so well, especially through the phones in our pockets?
- And, finally, what actually helps a person change their mind, including you?

The examples come from India and from around the world, because these mechanisms are human rather than national. You will read about a city that queued to feed milk to statues, a crowd that drank from a sewage-laden creek because they believed it had turned sweet, a famous chairman who described his fraud as riding a tiger, an eyewitness who was certain and wrong, and a detective writer who believed in fairies. You will also meet a number of ordinary people in ordinary situations. Where those everyday scenes are composites, built to illustrate a pattern rather than to report a specific person's life, the text says so.

A note on evidence and humility

A book about how people fool themselves had better not fool its readers. Psychology has gone through a painful and healthy period of self-examination over the past fifteen years, often called the **replication crisis**. When researchers tried to repeat some famous studies with larger samples and stricter methods, a number of results shrank, wobbled or disappeared. Others held up well.

This book takes that history seriously. You will find, in most chapters, a box headed **What the research does not say**. These boxes flag findings that are disputed, that have failed to replicate cleanly, that vary between cultures, or that are often exaggerated in popular retellings. The aim is not to make you distrust all psychology. It is to model the very habit the book is trying to teach: holding a belief with a strength that matches the evidence for it.

A related point. Many classic studies were run on small numbers of American university students. People in Pune, Patna or Palakkad are not always going to behave the same way, and in at least one famous case you will read about, a study with Indian participants helped show that an "error" once treated as universal was in fact shaped by culture. Where cultural differences are known, the book mentions them. Where they are unknown, it tries not to pretend.

How to use this book

Each chapter follows a similar shape:

1. **An opening scene**, real or composite, that shows the problem in action.
2. **The psychology**, explained in plain language with the key studies behind it.
3. **A case study**, documented and sourced, from India or abroad.
4. **What the research does not say**, the limits and disputes.
5. **A reflection exercise**, short enough to do in ten or fifteen minutes with a notebook.
6. **Key takeaways**, for review.

The exercises matter more than they might seem. Self-deception is invisible from the inside; that is what makes it self-deception. Reading about confirmation bias does very little, on its own, to reduce it. Several studies suggest that people can learn all about biases and still believe themselves to be less biased than everyone else. Writing things down, testing predictions and asking specific questions of yourself are among the few ways to catch the press secretary in the act.

You can read the chapters in order, since each builds on the last, or you can dip in where your interest lies. University students may

find the notes at the end useful as a map to the original research. Readers interested mainly in misinformation might start with Chapters 6 and 7, then return to the earlier chapters to see why those stories land so well.

One request before you begin. As you read about people who believed strange or harmful things, you will feel the pull of a familiar thought: *I would never fall for that.* Notice the thought. Hold it lightly. Almost everyone in this book thought the same.

The goal is not to make you suspicious of everyone else. It is to make you a slightly more honest witness to your own mind, and a kinder, more effective companion to the people you disagree with. Being wrong is not a character flaw. It is the normal condition of a mind trying to make sense of a complicated world. Refusing ever to find out is the real problem.

Let us begin with the feeling that drives so much of it: the need to be right.

01

CHAPTER 1

The Need to Be Right

> *The first step in any self-deception is not a lie. It is a small, reasonable-sounding refusal to look.*

The winter the world did not end

In the autumn of 1954, a woman in a suburb of Chicago began receiving messages that she believed came from beings on another planet. She wrote them down through what she described as automatic writing. The messages warned that a great flood would destroy much of North America before dawn on 21 December. A small group of true believers would be rescued by a flying saucer.

The woman, Dorothy Martin, was called “Marian Keech” in the famous account that made her story known. She gathered a circle of followers. Some of them were not marginal figures. One was a physician who lost his university job over his involvement. Several quit their jobs, gave away possessions and said goodbye to family. They took the prophecy seriously enough to remove metal from

their clothing, because the messages said metal would be unsafe aboard the craft.

Among the group, unknown to the believers, were several observers. The psychologist Leon Festinger and his colleagues Henry Riecken and Stanley Schachter had read about the prophecy in a newspaper and saw an unusual opportunity. Here was a group making a clear, testable prediction with a precise date. What would happen to their beliefs when the date came and went?

Midnight arrived. No spaceman knocked. The group sat in silence as the hours passed. Some wept. Then, a little before five in the morning, Martin received a new message: the small group's faith had spread so much light that God had spared the world from destruction.

What happened next is the part that made the case famous. Before the failed prophecy, the group had been fairly reluctant to seek publicity. Afterwards, according to the observers, several members began actively calling newspapers and trying to persuade others. The disconfirmation did not end the belief for the most committed members. In some ways it intensified their efforts to share it.

Festinger and his colleagues told the story in a 1956 book, *When Prophecy Fails*. It became the founding case study for one of the most influential ideas in social psychology: **cognitive dissonance**.

The discomfort of contradiction

Festinger's insight was simple to state. When people hold two thoughts that clash, or when their behaviour clashes with their beliefs, they experience an unpleasant tension. He called it dissonance. And like hunger or thirst, dissonance motivates us to make it go away.

There are several ways to reduce it. You can change your behaviour. You can change one of the beliefs. You can add new thoughts that make the clash seem smaller. Or you can avoid information that would sharpen it in the first place.

In principle, the rational option is often to change the belief that the evidence contradicts. In practice, Festinger argued, people choose whichever route is least costly to them. For Martin's followers, changing their belief about the prophecy meant admitting that they had given up jobs, savings and relationships for nothing. It meant facing the people who had warned them. It meant seeing themselves as foolish. Reinterpreting the failure as a success, and seeking others who might agree, was psychologically cheaper.

Consider how familiar the pattern is in smaller forms:

- A man buys an expensive car, then finds himself reading glowing reviews of that model and skipping the critical ones.
- A student who has spent four years preparing for a highly competitive exam, and has failed several attempts, explains each failure as bad luck, an unfair paper or a single weak subject, and resists any conversation about other careers.
- An investor who bought a stock at its peak insists it is "a long-term holding", a label he never used before the price fell.
- A family that spent a fortune on a lavish wedding becomes the loudest defender of lavish weddings.

In each case the person is not just believing something. They have **invested** in it: money, time, public commitment, a sense of who they are. The more they have invested, the more painful it is to discover that the investment was a mistake, and the more effort the mind puts into discovering that it was not.

One dollar versus twenty

Festinger did not rely only on observation. In a classic 1959 experiment with James Carlsmith, students at Stanford University spent an hour doing deliberately boring tasks, such as turning pegs on a board a quarter turn at a time. Afterwards, the experimenter asked each student to tell the next participant that the task had been interesting and enjoyable. Some were paid one dollar for this small lie. Others were paid twenty dollars, a substantial sum in the 1950s.

Later, the students were asked how much they had actually enjoyed the task. Those paid twenty dollars rated it as dull, which it was. Those paid one dollar rated it as more enjoyable.

Why would less money produce more liking? Festinger and Carlsmith's explanation was that the twenty-dollar students had a good external reason for the lie. "I said it because I was paid well." No dissonance. The one-dollar students had no such excuse. "I told someone a boring task was fun, and I did it for almost nothing." The easiest way to reduce that uncomfortable thought was to decide, at some level, that the task had not been so boring after all.

The study is a vivid demonstration of a broader principle: **when we cannot justify our behaviour by pointing outside ourselves, we tend to adjust our beliefs to fit what we have already done.** We do not only act on our beliefs. We also believe in line with our actions.

Related research supports the same idea. In 1959, Elliot Aronson and Judson Mills found that women who went through an embarrassing initiation to join a discussion group later rated the group, which was designed to be dull, as more interesting than women who had a mild initiation or none. Having suffered to get in, they had reason to believe it was worth it. In a 1968 field study at a Canadian racetrack, Robert Knox and James Inkster found that people who had just placed a bet on a horse were more confident in its

chances than people who were still standing in the queue to place their bet. The horse had not changed. The bettor's commitment had.

The ego in the middle

Later researchers, including Aronson himself, refined Festinger's theory in an important way. Not every inconsistency bothers us equally. People cheerfully live with dozens of minor contradictions. What reliably produces dissonance, Aronson argued, is a clash that threatens our view of ourselves as **competent, moral and sensible** people.

That refinement matters for this book, because it explains why self-deception tends to cluster around the self. "I made a mistake" is a manageable thought. "I am the sort of person who makes stupid mistakes" is not. "I acted badly" is uncomfortable. "I am a bad person" is intolerable. The mind works hardest to protect whatever is closest to the core of identity.

The social psychologists Carol Tavris and Elliot Aronson described this process in their book *Mistakes Were Made (But Not by Me)* with the image of a pyramid. Two people stand at the top, both undecided about some small choice: cheating on an exam, say, or cutting a corner at work. One goes one way, the other goes the other way. Each then justifies their choice, and the justification nudges them a little further down their side of the pyramid. After a series of small steps, each justified, they end up at the bottom far apart, each convinced that the other is obviously wrong and that their own position is the only reasonable one. Neither remembers how close they once stood.

The pyramid is a metaphor, not a measured finding, but it captures something important. **Self-justification is usually incremental.**

Nobody wakes up and decides to become a fraud, a fanatic or a person who ignores evidence. People take one small step, justify it, and then the next step feels smaller because of the justification for the first.

CASE STUDY: "LIKE RIDING A TIGER" — THE SATYAM CONFESSION

On 7 January 2009, B. Ramalinga Raju, the founder and chairman of Satyam Computer Services, then one of India's most celebrated IT companies, sent a letter to his board and to the stock market regulator. In it he admitted that the company's accounts had been falsified for years. The cash and bank balances on the books had been inflated by more than five thousand crore rupees, money that did not exist.

The scale was shocking. So was the source. Satyam had won awards for corporate governance. Raju had been held up as a model of the new Indian entrepreneur.

What interests us here is the shape of the story the letter told. As reported in the press, Raju described the fraud as starting small: a gap between actual and reported operating profit that grew year after year. In one widely quoted line, he wrote that it was *like riding a tiger, not knowing how to get off without being eaten.* He described a last attempt, through a proposed acquisition of two real-estate companies linked to his family, to replace fictitious assets with real ones. When the company's independent directors and investors objected and the deal collapsed, the hole could no longer be covered.

Whatever one makes of Raju's moral responsibility, and a court later convicted him, the letter is a remarkable portrait of the pyramid of self-justification in action. Each year, the previous year's inflated numbers made honesty more costly. Admitting the truth in year three would have meant admitting to years one and two as well. So the falsification continued, presumably accompanied by a story in which the problem was temporary, the business was fundamentally sound, and a solution was just around the corner.

> There is a final twist. In 2010, Raju's lawyers told the court that he had never written the confession letter, and the defence argued it was not genuine. A court ultimately convicted him in 2015. We cannot know what Raju privately believed at each stage. But the case shows how a person can travel, one justified step at a time, from a single bad quarter to one of the largest corporate frauds in Indian history, and how the story people tell about their own conduct can keep shifting even after the facts are public.

Why being right feels so good

Dissonance theory explains why being wrong feels bad. But there is also a positive pull towards being right that deserves its own attention.

Being right is pleasurable. Anyone who has won an argument at the dinner table, or seen a prediction come true, knows the small warm glow that follows. Being right also carries social rewards. In most groups, the person who is usually correct gains respect and influence. In many Indian families, being seen as the one who "knows" things, whether about medicine, money, cricket or politics, carries real status. Admitting error can feel like handing that status to someone else.

Some researchers have suggested that the feeling of certainty is itself something like an emotion, a sensation that arrives with a belief rather than a conclusion reached by checking it. The neurologist Robert Burton made this argument in his book *On Being Certain*. Whether or not one accepts that framing, it points to a useful practical fact: **how sure you feel is a poor guide to how likely you are to be right.** Certainty is a feeling the mind produces. It can be produced by good evidence, but it can also be produced by repetition, by group agreement, by the desire to be done with the question, or simply by having said something out loud.

Public commitment strengthens this effect. Once we have stated a position in front of others, especially online where the statement is preserved, changing it means not just updating a belief but contradicting our past self in public. In a well-known 1955 study, Morton Deutsch and Harold Gerard found that people who had written down their judgements and signed them were less likely to change their answers in the face of group pressure than people who had kept their judgements private. Commitment had made them consistent, even when consistency was not what accuracy required.

The lesson is not that we should never commit to anything. Commitments make us reliable and allow others to depend on us. The lesson is that **the same commitment that makes us dependable can also make us defensive**, and it helps to know which is happening.

The two-sided mirror

One of the most striking features of self-justification is how differently it operates when we look at ourselves and when we look at others.

When I arrive late, it is because the traffic was impossible, the auto driver took a wrong turn and my previous meeting overran. When you arrive late, it is because you are disorganised and do not respect my time. When I lose my temper, I was provoked. When you lose yours, you have a temper problem. When I change my opinion, I have learned something. When you change yours, you are a flip-flopper.

This asymmetry has been studied extensively. Psychologists have long observed what they call the **actor–observer asymmetry**: a tendency to explain our own behaviour in terms of circumstances and other people's behaviour in terms of their character. A related tendency, the **self-serving bias**, describes how we often take credit

for success (“I prepared well”) while blaming failure on outside factors (“the questions were unfair”).

We will look more closely in the next chapter at how these patterns connect to identity, and at evidence that their strength varies across cultures. For now, notice how neatly they serve the need to be right. If my mistakes are the product of circumstances, then they say nothing about my judgement. My self-image survives intact. And if your mistakes reveal your character, then I have good reason to discount your opinions, which means I need not take your criticism of me too seriously either.

Small doses, daily

It would be wrong to finish this chapter with the impression that the need to be right only matters in dramatic cases like doomsday cults and corporate frauds. Its most common effects are small and constant.

Consider a composite but familiar scene. Priya, a project manager in Bengaluru, recommended a software vendor to her company. Six months later, the vendor is missing deadlines. Her team complains. When her manager asks for a review, Priya finds herself writing a report that emphasises the vendor’s recent improvements, the difficulty of switching mid-project and the fact that every vendor has teething problems. All of these points are true. She does not lie. But she does not volunteer that she ignored two warning signs during the selection process, and she spends more time searching for reasons to stay than for reasons to leave.

If someone accused Priya of dishonesty, she would be genuinely offended, and with some justice. What she is doing is something subtler. She is acting as her own defence lawyer, collecting the arguments that support a conclusion she needs to be true, and she is

doing it so naturally that she does not notice the role she has taken on.

Most of the time, this does little harm. The stakes are low, the errors are small, and reality corrects us eventually. But the same pattern, scaled up and repeated, explains a great deal of human folly: the doctor who cannot accept that a treatment he has used for decades is harmful, the investigator who becomes certain of a suspect and stops looking for others, the government that keeps a failing policy because abandoning it would mean admitting it had failed.

WHAT THE RESEARCH DOES NOT SAY

- **Dissonance is not a universal law with precise predictions.** The theory has been debated for decades. Rival explanations, such as Daryl Bem's self-perception theory, account for some of the same results without invoking an uncomfortable inner state. Modern researchers generally see dissonance as a real and useful idea, but they argue about when it occurs and how strong it is.
- **Some classic studies are small and old.** The Festinger and Carlsmith experiment involved a modest number of students. Its central idea has been supported in many later studies using different methods, but the specific numbers from any single 1950s experiment should not be treated as precise.
- ***When Prophecy Fails* is a story, not a controlled experiment.** The observers were part of the group, and the book itself acknowledges that they may have influenced it. Later scholars have questioned how cleanly the case fits the theory, and studies of other failed prophecies show that many groups do *not* intensify recruitment; some simply dissolve. The story illustrates dissonance; it does not prove that failed predictions always strengthen belief.
- **Commitment does not always make people defensive.** People change their minds frequently, especially when the issue is not tied to their identity and when they can do so without losing face. That is part of the hopeful message of Chapter 8.

Bridge to the next chapter

The need to be right is strongest, this chapter has suggested, when being wrong would threaten our sense of being competent, moral and sensible people. But for many beliefs the threat goes deeper. Some beliefs are not just things we hold; they are part of who we are and of the groups we belong to. When a belief becomes part of identity, an argument about facts turns into an argument about loy-

alty, and the stakes of changing one's mind rise sharply. That is where we turn next.

REFLECTION EXERCISE: THE COST OF BEING WRONG

Take ten minutes and a notebook.

1. **Choose one belief you hold firmly** about something practical: a diet, a parenting approach, an investment, a career choice, a political policy. Write it down in one sentence.
2. **List what you have invested in it.** Money? Time? Public statements? Arguments with family members? A part of your self-image?
3. **Imagine conclusive evidence that you were wrong.** Not partial, not debatable, but conclusive. Write down, honestly, what you would lose. Who would you have to admit it to? What would it say about your judgement?
4. **Now ask:** is the strength of your belief based mainly on the evidence for it, or partly on the cost of giving it up? There is no need to change your mind. The aim is only to separate the two.
5. **Finally, write down one piece of evidence that would genuinely change your mind.** If you cannot think of any, notice that too.

KEY TAKEAWAYS

- When our beliefs clash with each other or with our behaviour, we feel an unpleasant tension called **cognitive dissonance**, and we are motivated to reduce it.
- We often reduce dissonance by changing our beliefs to fit what we have already done, rather than the other way round.
- Dissonance is strongest when a contradiction threatens our view of ourselves as **competent, moral and sensible**.
- Self-justification usually works in **small steps**; each justified step makes the next one easier.
- **Feeling certain is not the same as being right.** Certainty can be produced by commitment, repetition and social reward as well as by evidence.
- We tend to explain our own mistakes by circumstances and other people's by character, a pattern that protects our self-image.

02

CHAPTER 2

How Beliefs Protect Identity

> *We do not simply have beliefs. Some of our beliefs have us.*

The argument that was never about the facts

Here is a scene many Indian families will recognise. It is a composite, but the shape of it is real.

Ravi's grandmother has been unwell. His cousin, a junior doctor in Lucknow, wants her to start a course of prescribed medication. Ravi's uncle, who has run a successful business for thirty years and has always managed his own health with home remedies, insists that the tablets will "damage her liver" and that a herbal preparation from a trusted vaidya is safer. The family WhatsApp group fills with forwarded articles. Each side posts research. Each side finds the other's research unconvincing.

Ravi notices something odd. The argument is ostensibly about pharmacology, but no one is behaving as if it is. His uncle speaks with the heat of a man defending his honour. His cousin's replies

carry a tone of exasperation that has less to do with the liver than with years of being treated as the family's junior. When an aunt suggests consulting a second doctor, both sides hear it as taking the other side.

What is happening here? The facts matter, and the grandmother's health is genuinely at stake. But the argument has been captured by something else. For the uncle, the herbal approach is tied to his independence, his respect for tradition and his status as the family elder whose judgement is trusted. For the cousin, modern medicine is tied to her professional identity and her hard-won authority. Neither can concede the factual point without seeming to concede something much larger about who they are.

This chapter is about that capture: how beliefs become fused with identity, and why that fusion makes reasoning so much harder.

Beliefs as badges

We usually think of beliefs as private judgements about how the world is. But many beliefs also perform a social function. They signal which group we belong to, which values we hold and which people we stand with.

Some beliefs do this obviously: religious creeds, political affiliations, loyalty to a cricket team. Others do it more quietly. In different communities and at different times, opinions about vaccines, diets, climate change, genetically modified crops, traditional medicine, the right age to marry, or the merits of a particular film star have all functioned as markers of identity. Once a belief becomes a badge, holding it is no longer just a matter of evidence. It is a way of belonging.

The Yale law professor Dan Kahan and his colleagues have studied this phenomenon under the name **identity-protective cognition**.

Their argument is uncomfortable but compelling. For any single individual, being wrong about a large public question like climate science has almost no personal cost; one person's opinion has negligible effect on global temperatures. But being out of step with one's community on such a question can be very costly: lost friendships, awkward family dinners, a reputation as a traitor or a fool. Seen this way, it can be *individually rational*, in a narrow sense, to believe whatever your group believes, even if the result is collectively disastrous.

In one widely cited 2012 study, Kahan's team examined how Americans' concern about climate change related to their scientific literacy and numeracy. A common assumption was that people who doubted climate science simply did not understand it, and that more education would bring them round. The data told a different story. Among people with egalitarian, community-oriented values, higher science literacy went with more concern. Among people with hierarchical, individualist values, higher science literacy went with slightly *less* concern. Overall, the most scientifically literate people were the most polarised. Knowledge was not drawing people together. It seemed to be helping them defend the position their group already held.

We will return to the role of intelligence in Chapter 5. For now, the point is this: **when a belief is tied to identity, more information does not automatically produce more agreement.** It can simply produce better-armed disagreement.

Why an attack on a belief feels like an attack on you

There is a reason that criticising someone's belief can produce reactions out of all proportion to the topic. If a belief is part of my identity, then your criticism of it is, in a real sense, criticism of me. It implies that I am foolish, or disloyal, or bad. My mind responds as

it would to any threat to my standing: by defending, counter-attacking or withdrawing.

This is why the usual strategy of correcting someone, laying out the facts clearly and waiting for them to agree, so often fails in emotionally charged areas. The facts may be perfectly good. But they arrive packaged with an implicit message: *you were wrong, and people like you are wrong.* The listener rejects the package, and the facts go with it.

Fortunately, research also suggests something hopeful. If the threat to identity can be reduced, people sometimes become noticeably more open to unwelcome evidence.

In a set of experiments published in 2000, Geoffrey Cohen, Joshua Aronson and Claude Steele asked people to read evidence that challenged their views on a contested issue, such as capital punishment. Before doing so, some participants completed a **self-affirmation** exercise: they wrote about a value that was important to them and unrelated to the issue, such as their relationships with friends or family, or their sense of humour. Those who had affirmed their values evaluated the challenging evidence somewhat more even-handedly and were somewhat more likely to shift their views.

The idea behind self-affirmation is that our sense of self-worth rests on many pillars. If one pillar is threatened, we defend it fiercely. But if we are reminded of other pillars first, the threat to any single belief feels less catastrophic. We can afford to be wrong about this one thing, because we are still a good, competent person in many other ways.

There is an important caution here, which appears in the box at the end of this chapter: self-affirmation findings have proved more variable in later research than early studies suggested. But the core intuition is sound and is echoed in many traditions of conflict res-

olution. **People change their minds more readily when changing their mind does not require them to lose face.**

CASE STUDY: THE DOCTOR WHO ASKED DOCTORS TO WASH THEIR HANDS

In the 1840s, the Vienna General Hospital ran two maternity clinics. Women were admitted to them on alternate days. The First Clinic was staffed by doctors and medical students; the Second by midwives. The death rate from puerperal fever, also called childbed fever, was dramatically higher in the First Clinic. In some years it was several times higher. Women knew this, and some begged not to be admitted on the First Clinic's days. A few preferred to give birth in the street on the way.

Ignaz Semmelweis, a young Hungarian physician working in the First Clinic, set out to understand why. He tested and eliminated several explanations. Then, in 1847, a colleague, Jakob Kolletschka, died after being cut with a scalpel during an autopsy. His symptoms closely resembled those of the dying mothers.

Semmelweis drew a connection. The doctors and students in the First Clinic often came straight from performing autopsies to examining women in labour. The midwives did not do autopsies. Perhaps, he reasoned, the doctors were carrying "cadaverous particles" on their hands. He introduced a rule: wash hands in a chlorinated lime solution before examining patients. The death rate in his clinic fell sharply, to levels comparable with the midwives' clinic.

One might expect such a result to be welcomed. Instead, Semmelweis's ideas met strong resistance. Some of this resistance had reasonable grounds. Germ theory had not yet been developed, so his explanation lacked a mechanism that others could accept. He was slow to publish his findings formally, and he was, by many accounts, a combative communicator who attacked critics personally.

But part of the resistance seems to have come from somewhere else. Semmelweis's conclusion carried a terrible implication: that doctors themselves, respected men who considered their profes-

sion a noble calling, had been killing their patients. Accepting the hypothesis meant accepting a share of responsibility for many deaths. For physicians whose identity rested on being healers and gentlemen, this was not a neutral factual claim. It was an accusation.

Semmelweis's career in Vienna faltered. He later returned to Hungary, where he continued his work, but his professional and personal life deteriorated. He died in 1865 in an asylum, at the age of forty-seven. Within a few decades, the work of Louis Pasteur and Joseph Lister established germ theory and antiseptic practice, vindicating the essence of what Semmelweis had observed.

The phrase "Semmelweis reflex" is now sometimes used to describe the reflexive rejection of new evidence that contradicts established norms. It is a popular term rather than a scientific finding, and the real story is more complicated than a simple tale of a hero and his blind opponents. But the case remains a powerful example of how evidence can be resisted when accepting it would require a group to see itself in a deeply unflattering light.

Your mistakes, my mistakes and culture

In Chapter 1, we noted a familiar asymmetry: we tend to explain our own failures by circumstances and other people's failures by their character. This connects directly to identity. If my failures are situational, they say nothing about who I am. If your failures reveal your character, I may feel justified in thinking less of you, and of your group.

Psychologists studied a version of this pattern for decades under the name **fundamental attribution error**: the tendency, when explaining other people's behaviour, to overestimate the role of their personality and underestimate the role of the situation. In a classic 1967 study by Edward Jones and Victor Harris, American students read essays that either supported or criticised Fidel Castro's gov-

ernment in Cuba. Some were told that the essay writers had freely chosen their position. Others were told that the writers had been *assigned* their position by a debate coach. Even when readers knew the position had been assigned, they still tended to assume that the writer personally believed what they had written. The situation, an instruction to argue one side, was obvious, yet it was not given enough weight.

For a long time, many textbooks treated this tendency as a universal feature of human thinking. That is where India enters the story.

In 1984, the cultural psychologist Joan Miller published a study comparing how Hindu Indians and Americans explained everyday behaviour. She asked participants of different ages to describe things that people they knew had recently done, good and bad, and to explain why. She then coded the explanations. American adults were much more likely to explain behaviour in terms of general personality traits: *he is irresponsible*, *she is generous*. Hindu Indian adults were much more likely to refer to context: social roles, obligations, circumstances and relationships. Interestingly, the difference was much smaller among young children in both cultures and grew with age, suggesting that it was learned rather than innate.

Later research, including work comparing East Asian and Western participants, supported the broader conclusion: the tendency to attribute behaviour to character is real, but its strength depends partly on culture. Similarly, a large 2004 meta-analysis of the self-serving bias led by Amy Mezulis found the bias across many groups but noticeably weaker in several Asian samples than in American ones.

This is a useful corrective, and not only for scientific reasons. It reminds us that the specific ways our minds protect identity are shaped by what our culture treats as the core of identity. In settings that emphasise individual character and achievement, people may

be especially motivated to protect their image as competent individuals. In settings that emphasise family, community and social roles, the identities that need defending may be different: family honour, the reputation of one's caste, community or institution, the standing of one's elders. The protective machinery is universal. What it is guarding varies.

The tribes we carry

Identity-protective reasoning becomes especially visible around group loyalties. Consider a few ordinary examples.

Sport. Ask fans of two rival teams to watch the same disputed decision, a run-out given or not given, a catch that may have touched the ground, and you will often get two confident and opposite readings of the same replay. We will see in the next chapter that this pattern has been studied experimentally since the 1950s.

Profession. Engineers, doctors, lawyers, journalists and civil servants each develop shared beliefs about what good practice looks like. These shared beliefs are mostly valuable. But they can also make it hard for members of a profession to accept evidence that their established practice is wrong, as the Semmelweis story shows.

Region and language. Debates about language policy or regional achievements can quickly become identity contests in which statistics are used less to learn than to win.

Politics. This is the obvious case, and we will treat it at length in Chapter 6. For now, note only that party loyalty can be powerful enough to change how people evaluate the same policy depending on who proposes it.

In each case, the belief that protects identity often contains a good deal of truth. Fans are right that referees make mistakes. Profes-

sionals are right that their training gives them expertise. The problem is not that identity-linked beliefs are always false. It is that **identity makes us selective about which truths we are willing to see**.

When the self is the story

There is one identity that matters more than any group: the story we tell about ourselves.

Psychologists such as Dan McAdams have studied the **life stories** people construct to make sense of who they are. These narratives are not fixed records. They are revised over time to give our lives coherence and meaning. A failed marriage becomes the experience that taught us what we really needed. A lost job becomes the push that led to a better career. These redemptive stories can be healthy and adaptive; people who find meaning in difficulty often cope better.

But the same storytelling capacity can also protect us from uncomfortable truths. If my story is that I am a generous person, I may struggle to see the times I was selfish. If my story is that I am a victim of unfair treatment, I may struggle to see my own contribution to conflicts. If my story is that I am a self-made success, I may underrate the luck, family support or social advantages that helped me.

Consider another composite. Arjun, a successful entrepreneur in Gurugram, often speaks at events about how he built his company "from nothing" through hard work and risk-taking. Both are true. He did work hard, and he did take risks. What his story leaves out is the substantial loan from his father that kept the company alive during its second year, the network of family friends who became early clients and the elite education that made investors take him

seriously. Arjun is not consciously hiding these facts. If asked directly, he would acknowledge them. But they do not fit the story, so they rarely come to mind.

This is self-deception at its most ordinary. It does not involve false beliefs so much as **selective emphasis**. The facts that support our self-image are bright and close; the facts that complicate it are dim and distant. Over time, the story feels like the whole truth.

WHAT THE RESEARCH DOES NOT SAY

- **Identity does not make everyone impervious to evidence.** Most people update their factual beliefs when given clear corrections, even on politically charged topics, as Chapter 7 discusses. Identity seems to affect how much people resist, and which conclusions they draw from facts, more than whether they register facts at all.
- **The climate–literacy finding is specific to its context.** Kahan's results come from American samples on American political divisions. They illustrate a mechanism; they do not show that more education always increases polarisation everywhere.
- **Self-affirmation effects are inconsistent.** Some later studies and large replication efforts found smaller or no effects of brief self-affirmation exercises, both on openness to information and in other areas such as education. It is best treated as a promising idea with mixed evidence, not a reliable technique.
- **The "fundamental attribution error" is not fundamental in the sense of universal.** As Miller's research showed, its strength varies with culture and develops with age. Some researchers also argue that early studies overstated how badly people ignore situations.
- **The Semmelweis story is often simplified.** His own communication style, the lack of a germ theory and his delayed publication all contributed to his rejection. The case shows identity resisting evidence, but it is not only that.

Bridge to the next chapter

If beliefs can be tied to identity, and identity is something we defend, then we should expect the defence to begin early: not just in how we argue about evidence, but in which evidence we notice in the first place. The next chapter looks at the quiet, often invisible ways the mind filters information before we are even aware of having made a choice.

REFLECTION EXERCISE: MAPPING YOUR BADGES

1. **List five beliefs you hold that other people in your family, community or social circle would be surprised or upset to hear you abandon.** These might be about religion, politics, food, health, education, a sports team or a public figure.
2. **For each belief, write down who you would disappoint** if you changed your mind. Be specific. A parent? A group of friends? Your younger self?
3. **Pick the belief where the social cost is highest.** Ask yourself: if I had grown up in a different family or community, would I hold this belief as strongly? What would I believe instead?
4. **Try a short self-affirmation.** Write three or four sentences about a value that matters to you and has nothing to do with that belief: kindness, honesty, curiosity, loyalty to a friend. Then re-read the strongest argument you know *against* your belief. Notice whether it feels any different.
5. **Reflect:** what would it take for you to be able to change your mind on this question without feeling you had lost part of yourself?

KEY TAKEAWAYS

- Many beliefs function as **badges of identity**, signalling which groups we belong to and what values we hold.
- **Identity-protective cognition** means people may reason in ways that protect their standing in their group, even at the cost of accuracy.
- More knowledge does not automatically reduce disagreement on identity-linked issues; it can sometimes **arm people to defend** their existing positions.
- Criticism of an identity-linked belief is experienced as **criticism of the self**, which triggers defence rather than reflection.
- People are more open to changing their minds when doing so **does not require losing face**.
- The tendency to blame others' character and excuse our own behaviour is real, but its **strength varies by culture**, as research with Indian participants helped show.
- Our **life stories** can protect us from inconvenient truths through selective emphasis rather than outright falsehood.

03

CHAPTER 3

The Evidence We Choose to See

> *When we want to believe something, we ask, "Can I believe this?" When we don't, we ask, "Must I believe this?" The gap between those two questions is where much self-deception lives.*

The morning the gods drank milk

Before dawn on 21 September 1995, a worshipper at a temple in New Delhi offered a spoonful of milk to an idol of Lord Ganesha. As the spoon touched the idol's trunk, the milk appeared to vanish. Word spread. By mid-morning, the news had moved across Delhi and then across India by telephone, by word of mouth and through news reports. Within hours, people were reporting the same phenomenon at temples in Mumbai, Chennai and Kolkata, and among Hindu communities in Britain, the United States, Canada and elsewhere.

Queues formed outside temples. Milk sold out. In Delhi, according to reports, milk sales rose by more than thirty per cent. Offices emptied as people went to witness the miracle for themselves. The international press covered the event extensively.

Scientists and rationalists offered an explanation almost at once. A group of scientists tested the phenomenon by offering milk mixed with coloured dye. The milk, they found, was not vanishing. It was being drawn from the spoon by **surface tension and capillary action**, then running in a thin film down the front of the idol, often hidden by garlands, flowers and the base of the statue. Maharashtra's rationalist organisation, the Andhashraddha Nirmulan Samiti, demonstrated that the same effect could be produced with other materials and other liquids. Television footage showed milk pooling at the base of idols.

The scientific explanation did not end the queues. Many devotees continued to see a miracle. Some argued that the phenomenon had stopped too quickly, within about a day in most places, to be explained by ordinary physics. Others pointed to the sheer number of witnesses: surely so many people could not all be mistaken?

This book is not the place to judge anyone's faith, and that is not the point of this story. What is striking is something else: **thousands of people looked directly at the same physical event and saw what they expected to see.** The milk ran down the idol in plain view. But for those who approached with the expectation of a miracle, the vanishing at the spoon was the event, and everything that happened afterwards was background. For sceptics who approached expecting a trick, the run-off was the event. Expectation did not just shape interpretation. It shaped where people looked.

Looking for yes

In 1960, the British psychologist Peter Wason published a simple experiment that has become famous. He told participants that the number sequence 2, 4, 6 followed a rule he had in mind. Their task was to discover the rule by proposing other sequences of three numbers. After each proposal, he would say whether it fitted the rule. When they felt confident, they could announce their answer.

Most participants quickly formed a hypothesis, such as "numbers increasing by two." Then they tested it by proposing sequences like 8, 10, 12 or 20, 22, 24. Each time, Wason said yes, the sequence fitted. Feeling confident, they announced their rule. Many were wrong.

Wason's actual rule was simply "any three numbers in increasing order." Sequences like 1, 2, 3 or 5, 50, 500 would have fitted too. But participants rarely tried them, because they were testing for confirmation. They proposed examples they expected to fit their hypothesis, rather than examples that might show it was wrong. The most informative test, a sequence that their own rule predicted would fail, rarely occurred to them.

This tendency, to look for evidence that would confirm what we already think rather than evidence that might disconfirm it, is called **confirmation bias**. Later researchers have debated exactly how to interpret Wason's task. Some argue that testing positive cases is often a sensible strategy in the real world. But the broader pattern is robust and recognisable: we tend to search for, notice, remember and give weight to information that supports our existing views.

Confirmation bias is not one thing but a family of habits. It shows up in at least four places:

1. **What we search for.** We type questions into search engines in ways that presuppose our answer. *Why is this diet so effective?* produces different results from *Does this diet work?*
2. **What we notice.** In a busy information environment, items that match our expectations stand out; mismatches slide past.
3. **How we interpret.** Ambiguous evidence is read in the most favourable light for our existing view.
4. **What we remember.** Hits are recalled; misses are forgotten.

The last of these explains a great deal about why some practices survive despite weak evidence. Think of a family astrologer whose predictions are vague enough to fit many outcomes. When something he predicted seems to come true, the family remembers it for decades: *he said there would be a difficulty in the third month, and look what happened.* The many predictions that did not come true are rarely discussed. Over time, the memory of the successes creates the impression of a reliable track record.

Same game, different matches

In 1951, a rough American college football game between Princeton and Dartmouth produced injuries on both sides, including a broken nose for Princeton's star player. Afterwards, newspapers from each university blamed the other team. Two psychologists, Albert Hastorf of Dartmouth and Hadley Cantril of Princeton, saw an opportunity.

They showed a film of the game to students at both universities and asked them to note every rule violation they saw. The results, published in 1954 under the memorable title "They Saw a Game", were striking. Princeton students saw Dartmouth commit far more viola-

tions than Dartmouth students did. Dartmouth students saw a much more balanced game. Both groups had watched the same film.

Hastorf and Cantril concluded that there was, in an important sense, no single "game" that existed independently of the people watching it. Each group had experienced a different event, shaped by their loyalties.

Anyone who has watched an India–Pakistan cricket match with a mixed group of fans, or argued about a contested umpiring decision, will recognise this. Replays do not always settle these disputes. Fans watch the same frame and see the ball pitching in line or outside it, the edge or no edge. Some of this is the genuine ambiguity of the evidence. But the *direction* in which the ambiguity is resolved tends to follow loyalty.

Two questions, two standards

The psychologist Thomas Gilovich has described a useful way to think about motivated evaluation. When we encounter evidence for something we *want* to believe, we tend to ask ourselves, in effect: *Can I believe this*? Is there any reasonable basis for accepting it? The answer is usually yes, because almost any claim has some support.

When we encounter evidence for something we *do not* want to believe, we ask a different question: *Must I believe this*? Is the evidence so overwhelming that I have no choice? That is a much higher bar, and it is often not met, because almost any claim has some weaknesses.

Experiments support this asymmetry. In a 1992 study, Peter Ditto and David Lopez told participants they were taking a medical test for a fictional enzyme deficiency. Participants put a strip of test pa-

per in their saliva and waited for it to change colour. Some were told that a colour change meant they were healthy; others were told it meant they had a condition linked to future health problems. In fact, the paper never changed colour.

People who believed that *no change* meant good news accepted the result fairly quickly. People who believed that no change meant bad news waited longer, were more likely to retest and rated the test as less accurate. Faced with an unwelcome result, they did more checking. Faced with a welcome one, they were satisfied.

Notice that this is not stupidity or dishonesty. Retesting an unexpected result is often sensible. The problem is the asymmetry: **we check bad news thoroughly and good news hardly at all.** Over time, this means our beliefs are shaped more by what we want than by what is true, even though every individual step feels reasonable.

Two people, same evidence, further apart

One of the most influential studies of biased evaluation was published in 1979 by Charles Lord, Lee Ross and Mark Lepper at Stanford. They recruited students who either strongly supported or strongly opposed capital punishment. Each student read summaries of two fictional studies about whether the death penalty deterred murder. One study supported deterrence; the other did not. The studies were designed so that their methods were roughly comparable.

Supporters of capital punishment rated the pro-deterrence study as more convincing and better conducted, and found flaws in the anti-deterrence study. Opponents did the reverse. After reading the same mixed evidence, participants on both sides reported that their views had become *stronger*, not weaker.

If people evaluated evidence neutrally, mixed evidence should moderate strong views on both sides. Instead, in this study, it seemed to polarise them. Each side absorbed the supportive study and explained away the opposing one.

This finding has been enormously influential, and it captures something real. But it also has an important caveat, described in the box at the end of this chapter: later research suggests that what people *say* about becoming more extreme does not always match measurable changes in their beliefs. The asymmetric evaluation of evidence is well supported. The claim that mixed evidence always drives people further apart is less secure.

Not looking at all

The most effective way to avoid unwelcome evidence is never to encounter it. Researchers call this **selective exposure** and, in its more deliberate form, **information avoidance**.

A 2009 meta-analysis led by William Hart, combining dozens of studies, found that people have a moderate preference for information that agrees with their views over information that challenges them. The preference was stronger when people were more committed to their position and when the information related to important values. Interestingly, the preference weakened when people expected the challenging information to be useful for some practical goal.

Information avoidance is more familiar than we like to admit. People avoid medical tests when they fear the result. Investors check their portfolio more often when markets are rising than when they are falling; economists have called this the "ostrich effect". Students avoid checking exam results. People in troubled re-

lationships avoid the conversation that would bring the problem into the open.

Sometimes, avoiding information is reasonable. Not every piece of bad news is useful, and some uncertainty is easier to bear than some knowledge. But when we avoid information that would change our decisions, we are choosing comfort over accuracy, often without fully admitting the choice to ourselves.

Modern technology makes avoidance easier than ever. We choose which channels to follow, which groups to join and which people to mute. Recommendation algorithms learn what we engage with and show us more of it. The popular phrase "filter bubble" probably overstates the effect, as Chapter 6 explains; most people still encounter some opposing views. But the tools for avoidance are now always in our pocket.

CASE STUDY: THE MILK MIRACLE, REVISITED

Return to 21 September 1995 and look again at what happened through the lens of this chapter.

Searching for confirmation. People went to temples specifically to see whether the idols would drink, and with a spoon in hand they produced exactly the conditions under which capillary action would draw the milk away. Few people, even among the curious, ran the disconfirming test: offering milk to a non-sacred object of the same material, or watching carefully where the milk went. When rationalists did run those tests, they found that ordinary objects drank too.

Interpreting ambiguous evidence. The milk did leave the spoon. That part was real. The question was what it meant, and the answer people reached depended on what they believed beforehand.

Asymmetric scrutiny. Many believers accepted the miracle on the basis of their own observation and others' reports, the "Can I believe this?" standard. The scientific explanation, by contrast, was held to

the "Must I believe this?" standard, with objections such as the phenomenon's sudden stop being raised against it.

Social reinforcement. Seeing hundreds of others in the queue, hearing reports from other cities and countries, and watching it on the news all raised confidence. We will return to this social amplification in Chapter 6.

It is worth being careful here. Believing in a miracle is not irrational in the same way as, say, misreading a medical test, because for a believer the existence of divine action is part of their framework, not a hypothesis to be tested. The rationalists' demonstration did not prove that miracles are impossible. It showed that this particular event had a natural explanation. What the episode illustrates is not a defect peculiar to religious people. Secular audiences show exactly the same patterns when watching a magician, interpreting economic data or judging a cricket replay. The milk miracle is memorable because it happened on such a scale, and because the explanation was so simple once you looked in the right place.

Why does the mind work this way?

If confirmation bias leads to errors, why do we have it? Several explanations have been proposed, and they are not mutually exclusive.

Efficiency. Testing every belief against every possible counterexample would be exhausting. In a stable world, assuming that what we already know is roughly right is often a sensible default.

Social function. As discussed in Chapter 2, beliefs serve social purposes. A mind that readily abandoned its group's beliefs at the first sign of contrary evidence might pay social costs.

Argument. The cognitive scientists Hugo Mercier and Dan Sperber have argued that human reasoning evolved primarily for arguing with others: for producing justifications and evaluating other

people's arguments. On this view, it makes sense that we are good at finding arguments for our own position and good at spotting weaknesses in others' arguments, but not very good at spotting weaknesses in our own. In a group, the biases cancel out: each person attacks the others' arguments, and the best ones survive. Alone, the same tendencies leave us stuck in our own preferences. We return to this idea in Chapters 5 and 8.

Whatever the origins, the practical consequence is the same. **The evidence we see is not a random sample of the evidence that exists.** It is filtered by what we look for, what we notice, how we interpret it and what we remember. To see more clearly, we have to deliberately search for the evidence our habits would otherwise filter out.

WHAT THE RESEARCH DOES NOT SAY

- **Confirmation bias does not mean people ignore all contrary evidence.** People do update their beliefs, often in the right direction, when evidence is clear. The bias is a tilt, not a wall.
- **The "attitude polarisation" finding is weaker than often claimed.** In the 1979 Lord, Ross and Lepper study, participants *reported* that their views had strengthened. Later studies that measured beliefs before and after, rather than asking people how they had changed, often found less polarisation or none. Biased evaluation is well established; mixed evidence reliably driving people further apart is not.
- **Positive testing is not always an error.** Some researchers have shown that in many real-world situations, testing cases you expect to confirm your hypothesis is an efficient strategy. Wason's task was designed so that it would fail.
- **Selective exposure is moderate, not total.** Meta-analyses find a real preference for agreeable information, but people also consume plenty of disagreeable information, especially when it is useful to them.
- **The milk miracle is not evidence that believers are less rational than others.** It illustrates universal patterns that operate in every domain.

Bridge to the next chapter

So far, we have treated evidence as something that arrives from outside: a news report, a test result, a replay. But much of what we believe about our lives rests on a different kind of evidence, our own memories. Surely, we think, we can at least trust our recollection of what we saw, said and did. The next chapter explains why that trust is often misplaced.

REFLECTION EXERCISE: THE DISCONFIRMATION SEARCH

1. **Pick a belief you hold with moderate confidence.** It should be something factual: a health claim, a prediction about the economy, a judgement about a public figure or an opinion about a product.
2. **Write down the evidence you have for it.** Then ask: how did this evidence come to me? Did I search for it, did it find me, or did I simply absorb it from people around me?
3. **Run the "Must I?" test on your own side.** Apply to your supporting evidence the same level of scrutiny you would apply to evidence on the other side. Where does it come from? Is the sample large? Who benefits if I believe it?
4. **Search deliberately for the best disconfirming evidence.** Not the weakest version of the other side, but the strongest. Phrase your search neutrally: *Is it true that…?* rather than *Why is it true that…?*
5. **Write two sentences:** what you found, and whether your confidence has changed. A small change is still a change.

KEY TAKEAWAYS

- **Confirmation bias** shapes what we search for, what we notice, how we interpret ambiguous information and what we remember.
- We tend to test our beliefs by looking for **examples that fit** rather than examples that might show we are wrong.
- People with different loyalties can watch the same event and **see different things**.
- We apply an easy standard ("**Can** I believe this?") to welcome evidence and a strict one ("**Must** I believe this?") to unwelcome evidence.
- We often avoid information altogether when we fear what it might tell us.
- The evidence we see is a **filtered sample**. Seeing clearly requires deliberately seeking out what our habits screen out.

04

CHAPTER 4

Why Memory Is Not a Recording

> *Memory is less like a video archive and more like a storyteller who retells the tale a little differently every time, and swears it has never changed.*

The certain witness

In July 1984, a man broke into the apartment of Jennifer Thompson, a twenty-two-year-old college student in Burlington, North Carolina, and raped her. During the attack, Thompson made a deliberate decision. She would study her attacker's face, his voice, every detail she could, so that if she survived she could identify him.

She survived. She worked with police to create a composite sketch. Days later, she was shown a set of photographs and picked out a man named Ronald Cotton. Then she picked him out again in a physical line-up. In court, she identified him with complete confidence. Cotton was convicted and sentenced to life in prison.

Cotton insisted he was innocent. In prison, he encountered another inmate, Bobby Poole, who resembled him and who was rumoured to have committed the crime. At a retrial, Thompson was shown Poole. She said she had never seen him before. Cotton was convicted again.

In 1995, DNA testing became available. It showed that Cotton was not the attacker. The DNA matched Bobby Poole. Ronald Cotton had spent more than ten years in prison for a crime he did not commit.

Jennifer Thompson had not lied. She had tried, under terrible circumstances, to do everything right. Her confidence was genuine. Her memory of Cotton's face as the face of her attacker was vivid and sincere. It was also wrong.

Years later, Thompson and Cotton met. Remarkably, they became friends and wrote a book together, *Picking Cotton*, about their experience and about the dangers of eyewitness error. Thompson later described how, once she had picked Cotton's photograph, his face seemed to replace her attacker's face in her memory. The identification process had not just retrieved her memory. It had changed it.

How memory actually works

Most people imagine memory as something like a recording. An event happens, the brain stores it, and later we play it back. The recording may fade, or we may struggle to find it, but what we retrieve is, we assume, a copy of what happened.

Decades of research have shown this picture to be misleading. Memory is **reconstructive**. When we remember, we do not replay a stored file. We rebuild the experience from fragments: some details that were actually encoded, combined with general knowledge, ex-

pectations, later information and inferences about what must have happened. The result feels like a recording. It is better described as a reconstruction, and a reconstruction can be built differently each time.

One of the earliest demonstrations came from the British psychologist Frederic Bartlett in the 1930s. He asked English participants to read a Native American folk tale called "The War of the Ghosts" and then retell it on several occasions over weeks and months. The story contained elements that were unfamiliar to English readers. As participants retold it, the story changed in systematic ways. It became shorter. Unfamiliar details dropped out or were replaced by more familiar ones; a canoe became a boat, for example. Puzzling supernatural elements were rationalised. The story was gradually reshaped to fit the participants' own cultural expectations. Bartlett called these frameworks of expectation **schemas**.

Anyone who has listened to a family story told over many years at weddings and festivals has seen this process at work. An uncle's account of how the family moved to the city, how a grandparent survived Partition, or how a wedding nearly went wrong is polished with each telling. Dramatic details grow. Inconvenient ones fade. Different relatives develop different versions and each is certain theirs is correct. Usually no one is lying. The memories are being reconstructed, and reconstruction is shaped by what makes a good story and by what each teller needs the story to mean.

When words change what we saw

In the 1970s, the American psychologist Elizabeth Loftus began a long programme of research showing how easily memories can be altered by information received after an event.

In a famous 1974 study with John Palmer, participants watched films of car accidents and then answered questions about them. The key question asked how fast the cars were going when they "hit" each other. For some participants, the verb was changed: the cars "smashed", "collided", "bumped" or "contacted" each other. The single changed word affected the answers. People asked about cars that "smashed" gave higher speed estimates, on average, than people asked about cars that "contacted".

More striking was a follow-up. A week later, participants were asked whether they had seen any broken glass in the film. There had been no broken glass. Yet people who had earlier been asked about cars that "smashed" were more than twice as likely to report seeing broken glass as people who had been asked about cars that "hit". A single word, heard after the event, had added a detail to their memory of it.

Loftus and others went on to show this **misinformation effect** in many forms. Misleading questions, conversations with other witnesses and news reports can all introduce details that people later remember as part of the original event. Once incorporated, these details can feel just as real as genuine memories.

Remembering things that never happened

If memories can be altered, can entirely false memories be created? In the 1990s, Loftus and Jacqueline Pickrell tested this in a study that has become well known as the "lost in the mall" study.

With the help of family members, the researchers gave participants short descriptions of four events from their childhood. Three had really happened. One, being lost in a shopping mall at around the age of five and eventually rescued by an elderly person, had not.

Participants were asked to write about what they remembered of each event, and were interviewed about them later.

About a quarter of participants came to remember the false event, wholly or partly, sometimes adding details of their own. Later studies using similar methods produced false memories of events such as being hospitalised overnight, spilling punch at a wedding, or being attacked by an animal. A later meta-analysis of such "memory implantation" studies, led by Alan Scoboria, estimated that roughly a third of participants developed at least a partial false memory, though the figure depends heavily on how a false memory is defined.

These studies have been controversial, partly because of their implications for legal cases involving recovered memories of abuse, a topic discussed with caution in the box at the end of this chapter. But the basic finding is now widely accepted: **under the right conditions, people can come to remember, sincerely and in detail, events that did not happen.**

The vividness trap

Surely, you might think, some memories are too vivid to be wrong. Many people can recall exactly where they were when they heard about a major public event: the assassination of Indira Gandhi in 1984, the attacks on the World Trade Center in 2001, the Mumbai attacks of November 2008, India's World Cup victory in 2011. These are called **flashbulb memories**, because they feel as though a camera flash captured the moment in perfect detail.

Research suggests that these memories are not as reliable as they feel. On the morning after the space shuttle *Challenger* exploded in January 1986, the psychologist Ulric Neisser and his colleague Nicole Harsch asked university students to write down how they

had heard the news. About two and a half years later, they asked the same students again. Many of the later accounts differed substantially from what the students had written the day after. Some were completely different. Yet many students were highly confident in their later accounts. Some, when shown their original handwritten descriptions, were reluctant to accept them, insisting that their current memory must be correct.

After the September 2001 attacks, Jennifer Talarico and David Rubin asked students to record their memories of hearing the news and, as a comparison, a memory of an ordinary event from around the same time. When the students were tested again weeks and months later, their memories of the attacks had become less consistent at roughly the same rate as their ordinary memories. What differed was confidence. Students remained highly confident in their flashbulb memories, while their confidence in the everyday memories declined.

The lesson is uncomfortable. **Vividness and confidence are not reliable signs of accuracy.** Emotional events are memorable, and people rehearse them often, which keeps them vivid. But each rehearsal is a reconstruction, and the details can drift while the feeling of certainty remains.

Memory in the service of the self

Memory errors are not random. Like the other processes in this book, they tend to lean in a particular direction: towards a version of the past that makes us look good, that fits our current beliefs, and that tells a coherent story.

Flattering recall. In a 1996 study, Harry Bahrick and colleagues asked American university students to recall their high-school grades and then checked their answers against official records.

Students remembered their top grades very accurately. They were much less accurate about their lower grades, and their errors mostly went in one direction: grades were remembered as better than they had been.

Consistency with present views. People tend to remember their past opinions as being closer to their current opinions than they really were. When views change, memory quietly rewrites the past so that we seem to have been consistent, or so that the change seems smaller than it was.

Hindsight bias. After an event, we tend to believe we "knew it all along". In a classic 1975 study, Baruch Fischhoff found that people told the outcome of a historical event rated that outcome as more predictable than people who were not told. After an election, a stock crash or a cricket match, the result can feel inevitable, and our memory of how uncertain we were before the event fades. Hindsight bias makes us overconfident about our ability to predict and less willing to learn from surprises, because in retrospect nothing seems surprising.

Consider a composite example. Meena tells her friends she always had doubts about her former business partner. In fact, her diary from the early years of the partnership contains enthusiastic praise for him and notes about how lucky she was to have found him. Her current memory is not invented out of thin air; she did have some small doubts, which she now remembers as large. But her past has been reorganised around what she knows now. This makes the outcome feel less like a misjudgement and more like a risk she knowingly took.

CASE STUDY: TWO STORIES ABOUT ONE HELICOPTER

The newsreader's memory. In February 2015, Brian Williams, then the anchor of one of America's most-watched evening news pro-

grammes, apologised on air for a story he had told repeatedly. For years, he had described being aboard a military helicopter in Iraq in 2003 that was forced down after being hit by a rocket-propelled grenade. In fact, as soldiers involved pointed out, the helicopter that was hit was ahead of his; his helicopter arrived later. Williams was suspended for six months and lost his position as anchor.

Many people assumed Williams had simply lied to make himself look brave. That may be true. But memory researchers pointed out that the history of the story also fitted a pattern of gradual memory distortion. His early accounts were closer to the truth. Over years of retelling, the story seemed to drift, with the danger moving closer to him. Williams himself said he had conflated the two aircraft. We cannot know which explanation is correct, or whether both played a part. What the case shows is how hard it is, even for the person involved, to separate a lie from a memory that has been reconstructed many times in a flattering direction.

The eyewitness's memory. Return now to Jennifer Thompson. Her case illustrates almost every process described in this chapter.

- **Encoding under stress.** Extreme fear can narrow attention and impair memory for faces, despite a victim's best efforts.
- **Suggestion.** In the original photo line-up, Thompson hesitated between two photographs before choosing Cotton. Accounts of the case describe how feedback from police after the identification encouraged her. Research has since shown that such confirming feedback can greatly increase a witness's confidence, without making the memory more accurate.
- **Reconstruction.** After she chose Cotton's photograph, and especially after seeing him again in a line-up and at trial, his face became her memory of the attacker. Each viewing reinforced the reconstruction.
- **Confidence without accuracy.** By the time of the trial, she was certain. Juries tend to find confident witnesses highly persuasive.

The Innocence Project, an American organisation that uses DNA evidence to challenge wrongful convictions, reports that eyewitness

misidentification played a part in roughly seven in ten of the convictions later overturned by DNA evidence in the United States. Following cases like Cotton's, many police forces have changed their procedures: showing photographs one at a time, having line-ups run by officers who do not know which person is the suspect, recording a witness's confidence immediately, and avoiding feedback.

Indian courts have also long recognised the risks of identification evidence and the importance of how identification procedures are conducted. The principle is the same everywhere: **sincere certainty is not proof.**

Living with an unreliable narrator

If memory is reconstructive, should we distrust everything we remember? No. Most of our memories are accurate enough for daily life. We remember where we live, who our friends are and how to do our jobs. The gist of important events is usually preserved even when details drift.

The problem arises in three situations: when **details** matter (as in a courtroom), when **the self is at stake** (as in arguments about who said what, or who was to blame), and when **others' memories are mixed with ours** (through conversation, social media and repetition).

A few practical habits follow:

- **Write things down at the time.** A note made the same day is more reliable than a memory recalled a year later. Diaries, emails and photographs are external memories that do not drift.
- **Separate the gist from the details.** You may be right that an argument happened and that it hurt. You may be wrong about the exact words.

- **Be cautious with confidence.** Your certainty is information about your feeling, not about the past.
- **Notice when a memory flatters you.** If your recollection of a past event makes you look unusually wise, brave or blameless, it may be worth checking.
- **Allow others their versions.** In family disputes, couples' arguments and workplace conflicts, two sincere people can remember the same event differently. Neither need be lying.

WHAT THE RESEARCH DOES NOT SAY

- **Memory is not hopelessly unreliable.** Most memories are broadly accurate, especially for the gist of events. Research on errors focuses on errors, which can create a misleading impression.
- **False memory studies do not show that all recovered memories are false.** The implications of "lost in the mall" style studies for real cases of childhood abuse have been fiercely debated. Implanted memories of mildly unpleasant events in experiments are not the same as memories of trauma, and some real abuse is genuinely forgotten and later remembered. The evidence supports caution and careful procedures, not automatic disbelief.
- **The proportion of people who form false memories varies widely.** Estimates depend on the method and on how strictly a "false memory" is defined. Figures like "a quarter" or "a third" should not be quoted as fixed facts.
- **Eyewitness confidence is not useless.** More recent research suggests that a witness's confidence at the moment of a first, fair identification can be a meaningful indicator of accuracy. The problem is confidence that grows later through feedback and repetition.
- **We cannot know what Brian Williams believed.** The case is a useful illustration of how memory and self-presentation can blur together, not a diagnosis.

Bridge to the next chapter

By now, a reasonable reader might be thinking: these errors sound like the kind of thing that more careful, more educated, more intelligent people would avoid. Surely the solution is simply to be smarter. The next chapter examines that hope, and finds that intelligence can be a surprisingly double-edged tool.

REFLECTION EXERCISE: CHECKING YOUR OWN ARCHIVE

1. **Find a written record from your past.** An old diary entry, a set of messages, an email thread or a social media post from at least two years ago, ideally about a decision or an event that mattered to you.
2. **Before reading it, write down what you remember** about the situation: what happened, what you thought at the time and how you felt.
3. **Now read the record.** Compare it with your memory. What is different? What did you forget? What did you add?
4. **Look for the direction of the differences.** Did your memory make you look better, wiser or more consistent than the record shows?
5. **Reflect:** is there a current disagreement in your life, with a family member, a partner or a colleague, where you might be relying on a reconstructed memory? What would it mean to hold your version a little more lightly?

KEY TAKEAWAYS

- Memory is **reconstructive**: we rebuild the past from fragments, expectations and later information each time we remember.
- Information received **after** an event, even a single word, can change what we remember about it.
- Under some conditions, people can develop **detailed, sincere memories of events that never happened**.
- **Vividness and confidence** are not reliable signs that a memory is accurate.
- Memory errors tend to lean in a **self-serving direction**: towards a past that flatters us and fits our current views.
- **Hindsight bias** makes the past seem more predictable than it was, which can stop us from learning from mistakes.
- Practical protection comes from **records made at the time**, humility about details and generosity towards other people's versions.

05

CHAPTER 5

When Intelligence Becomes a Defence Mechanism

> *A clever mind is not a mind less likely to fool itself. It is a mind that can build a more elegant case for whatever it already wanted to believe.*

The creator of Sherlock Holmes and the fairies

In the summer of 1917, in the village of Cottingley in Yorkshire, England, two cousins borrowed a camera. Elsie Wright was sixteen. Frances Griffiths was nine. They went down to the stream at the bottom of the garden and came back with photographs that appeared to show Frances surrounded by tiny dancing fairies, and Elsie with a winged gnome.

For a few years, the photographs circulated quietly. Then they came to the attention of Edward Gardner, a prominent figure in the Theosophical movement, and through him to one of the most fam-

ous writers in the world: Sir Arthur Conan Doyle, the creator of Sherlock Holmes.

Doyle was a trained physician. He had created the most celebrated fictional rational detective in literature, a character synonymous with careful observation and logical deduction. He was also, by this point, a passionate believer in spiritualism, the conviction that the living could communicate with the dead. That belief had deepened in the years around the First World War, during which he lost his son Kingsley and other members of his family.

Doyle became convinced the photographs were genuine. In the Christmas 1920 issue of *The Strand Magazine*, the same magazine that had published Sherlock Holmes stories, he presented the fairy photographs to the public. In 1922 he published a book about them. He offered arguments: the girls were simple country children incapable of sophisticated fakery; photographic experts had found no evidence of double exposure; the fairies' poses showed motion.

Many people at the time pointed out obvious problems. The fairies looked remarkably like the fashionable illustrations of the day. Their hairstyles were modern. They did not seem to be in motion at all. Doyle responded to these objections with further arguments.

More than sixty years later, in 1983, the cousins, by then elderly women, admitted that the fairies had been cut out of paper, copied from illustrations in a popular children's book and held in place with hatpins. Frances maintained to the end of her life that the last of the five photographs was genuine. The rest were a childhood prank that had grown out of control and became too embarrassing to confess once famous people had staked their reputations on it.

How did a man trained in medicine, celebrated for creating the very model of sceptical reasoning, fall for paper fairies held up with pins?

The tempting answer is that Doyle's intelligence failed him. This chapter suggests a different answer: **Doyle's intelligence did not fail. It was working hard, and it was working for the wrong side.**

The myth of the immune expert

There is a comforting belief, common among educated people, that the errors described in this book are mainly problems for others: the less educated, the less intelligent, the people who forward implausible messages without thinking. If only those people knew more and thought harder, the reasoning goes, they would see the truth.

There is some truth in this. Knowledge matters. People who understand basic statistics, for example, are less likely to fall for certain kinds of numerical trickery. Some research has found that people who score higher on tests of analytical thinking are, on average, slightly better at telling true headlines from false ones, a finding we return to in Chapter 7.

But a growing body of research suggests that, when beliefs are tied to identity or strong desire, intelligence offers much less protection than we would hope. In some situations, it may even help us defend our existing beliefs more effectively.

The Canadian psychologist Keith Stanovich and his colleagues have studied what they call **myside bias**: the tendency to evaluate evidence, generate evidence and test hypotheses in a way that favours one's prior opinions. In a series of studies, they found that myside bias was surprisingly unrelated to measures of intelligence or cognitive ability. Smart people were roughly as likely as others to show it. Stanovich has argued that intelligence and rationality are different things. Intelligence is the capacity to process information effectively. Rationality includes the disposition to use that capacity in

the service of finding the truth rather than defending a position. A person can have a great deal of the first and rather little of the second, especially on topics they care about.

Better at arguing, not at doubting

Intelligence helps in many ways. It makes us better at learning, at solving problems and at constructing arguments. But constructing arguments is precisely the skill that motivated reasoning needs.

Return to the image of the inner press secretary from the introduction. A more capable press secretary is not more likely to tell the truth. It is more likely to produce a polished, persuasive explanation for whatever the leadership has already decided. If our conclusions are shaped by identity and desire, intelligence may simply make us better at justifying them.

Several studies support this concern, though the details are debated.

In a 2006 study, Charles Taber and Milton Lodge gave American participants arguments for and against policies such as gun control and affirmative action. Participants spent more time on arguments that contradicted their views, but used that time mainly to find reasons to reject them. Participants who were more politically knowledgeable showed some of the strongest biases. They knew more counter-arguments, and they used them.

In a widely discussed 2017 paper, Dan Kahan and colleagues asked Americans to interpret data from a fictional study. In one version, the data concerned whether a skin cream reduced rashes. In another, identical numbers concerned whether a ban on carrying concealed handguns reduced crime. Getting the right answer required careful attention to a two-by-two table, and people with higher numeracy generally did better. But on the gun-control ver-

sion, people's accuracy depended on whether the correct answer matched their political views. In Kahan's original data, the most numerate participants showed the largest gap. They were very good at getting the answer right when the answer suited them.

This is an alarming result, but it should be held with care. A large, pre-registered replication published in 2021 confirmed that people interpreted the data in a politically biased way, but did not clearly reproduce the finding that greater numeracy made the bias larger. The safest conclusion is that **numeracy did not protect people from motivated interpretation**, not necessarily that it made things worse.

The bias blind spot

There is another, more personal way that intelligence and education can backfire. Clever people often believe they are less biased than others, and that belief can make them less vigilant.

In 2002, Emily Pronin, Daniel Lin and Lee Ross described what they called the **bias blind spot**. They gave American participants descriptions of common biases, such as the self-serving bias, and asked how much they and the average person showed each one. Participants consistently rated themselves as less biased than the average person. When shown evidence that they had just displayed a bias, many still maintained that their own judgement had been objective.

Pronin's explanation is subtle. When we judge others, we look at their behaviour, and their behaviour often looks biased. When we judge ourselves, we look inward, at our thoughts and intentions, and our thoughts feel fair. We do not experience our biases as biases. We experience them as the truth. Our introspection reports "I

considered the evidence carefully", because from the inside that is what it felt like.

Later work by Richard West, Russell Meserve and Keith Stanovich found that the bias blind spot was not smaller in people with greater cognitive ability. If anything, in their samples, it was slightly larger. Clever people may have more evidence of their own competence to draw on, and so more reason to believe their reasoning is sound.

This creates a trap. The very knowledge described in a book like this one can be turned into a weapon. Once we know about confirmation bias, we can spot it in our opponents' arguments with great precision. We can name their errors. What we often cannot do is apply the same diagnosis to ourselves, because our own reasoning feels, from the inside, like simple clarity.

Experts and their hedgehogs

The political scientist Philip Tetlock spent nearly two decades, from the 1980s to the early 2000s, collecting predictions from hundreds of experts on political and economic events: whether a country's government would fall, whether a war would occur, whether an economy would grow. He then checked the predictions against what actually happened.

The overall results, published in 2005 in *Expert Political Judgment*, were humbling. On many questions, experts performed only modestly better than simple rules of thumb, and many were outperformed by them. But Tetlock found an important difference within the group, which he described using an old image from the philosopher Isaiah Berlin: the fox and the hedgehog. Hedgehogs know one big thing. They have a grand theory and apply it confidently to

everything. Foxes know many little things. They draw on different ideas, hold views tentatively and adjust when evidence changes.

Foxes were noticeably better forecasters than hedgehogs. Strikingly, the hedgehogs who were most famous, those most often quoted in the media, were often among the worst. Their confidence and their compelling single story made them excellent communicators and poor predictors. When their predictions failed, they were skilled at explaining why they had been almost right, or right for the wrong reasons, or would be right eventually.

Expertise, in other words, is not one thing. Knowledge helps. But the *style* in which knowledge is held matters at least as much: whether it is held as a fortress to be defended, or as a set of working hypotheses to be revised.

CASE STUDY: HOW A BRILLIANT MIND DEFENDS A MISTAKE

Return to Arthur Conan Doyle and look at how his intelligence was deployed.

He generated sophisticated arguments. Doyle did not simply assert that the fairies were real. He reasoned. He considered the girls' backgrounds, the camera, the expert opinions and the composition of the photographs. He anticipated objections and answered them.

He applied asymmetric scrutiny. Evidence for the photographs, such as a specialist's opinion that he could find no sign of double exposure, was accepted readily. Evidence against them, such as the fashionable hairstyles of the fairies and their resemblance to commercial illustrations, was explained away. This is the "Can I believe it?" versus "Must I believe it?" asymmetry from Chapter 3, carried out by a gifted arguer.

His belief was tied to his identity and his grief. By 1920, Doyle had become one of the most prominent public advocates of spiritualism. He had lost his son and other relatives. The existence of a hidden spiritual world was not a casual opinion for him; it was a source of

comfort and a cause to which he had committed his reputation. Fairies, if real, would support the larger worldview. Admitting he had been fooled by two schoolgirls would have damaged his standing as a spokesman for that worldview.

He resisted even contradictory testimony. The pattern appears elsewhere in Doyle's life. He became friends with the great escape artist Harry Houdini, who spent much of his later career exposing fraudulent mediums. Houdini insisted repeatedly that his feats were achieved through skill and trickery, not supernatural powers. Doyle came to believe that Houdini did in fact possess psychic powers and was concealing them. The friendship eventually collapsed over their disagreements about spiritualism. When even the man performing the trick tells you it is a trick, and you conclude that he is mistaken about himself, your reasoning is no longer testing a belief. It is protecting it.

None of this means Doyle was foolish. He was a highly intelligent and accomplished man who brought considerable skill to the task of defending what he wanted to be true. That is exactly the point.

A modern Indian echo

The same pattern can be seen in contexts much closer to home. Here is a composite that many readers will recognise.

Dr Suresh is a respected cardiologist in Hyderabad. He is careful and evidence-based in his medical practice; he reads journals, follows guidelines and is sceptical of unproven treatments his patients bring to him. Outside medicine, however, he is an enthusiastic investor in shares, guided largely by tips from a private group of fellow professionals. When his stocks rise, he credits his analysis. When they fall, he blames market manipulation, government policy or bad luck. He forwards articles that support his strategy and dismisses studies showing that most active traders underperform

simple index funds. “Those studies are about average investors,” he says. “We are not average.”

Dr Suresh’s intelligence is real. In his own field, it is accompanied by training, feedback and professional norms that force him to test his beliefs. In investing, those disciplines are absent, his ego is involved and he has a ready supply of sophisticated explanations. His expertise in one field gives him confidence in another where he has none.

Psychologists sometimes discuss a related pattern among highly accomplished people: success in one area can produce overconfidence in areas where one has no special knowledge. The point is not that doctors or engineers are especially gullible. It is that **competence in one domain does not transfer automatically to another**, and the confidence that comes with it often does.

Turning intelligence back on itself

If intelligence can be recruited to defend our beliefs, can it also be used to examine them? Research offers some hope, with conditions.

Reasoning works better in groups that disagree. Hugo Mercier and Dan Sperber’s argumentative theory of reasoning, mentioned in Chapter 3, suggests that our reasoning evolved for arguing with others. We are good at criticising other people’s arguments and poor at criticising our own. In a group where people genuinely disagree and evaluate each other’s arguments, these biases can cancel out. Some studies have found that small groups discussing logic problems reach the correct answer much more often than individuals do, because the person with the right answer can convince the others.

We are better critics of our own arguments when we don't know they are ours. In a clever 2016 study, Emmanuel Trouche, Petter Johansson, Lars Hall and Hugo Mercier asked people to answer reasoning problems and give reasons for their answers. Then, in a second stage, participants were shown other people's answers and arguments to evaluate. Through a sleight of hand, one of the "other people's" arguments was actually the participant's own. Many participants did not notice the switch, and around half of those rejected the argument they had themselves produced minutes earlier. When we think an argument belongs to someone else, we evaluate it more critically.

Actively open-minded thinking can be cultivated. Tetlock's later research on "superforecasters", ordinary people who produced unusually accurate forecasts in a large prediction tournament, found that they shared certain habits: they treated their beliefs as hypotheses, updated frequently in small steps, actively sought out disconfirming information and kept careful score of their predictions. These habits seemed to matter more than raw intelligence.

The practical lesson is not "be less intelligent". It is to **put your intelligence under supervision**. Use it as a prosecutor of your own views as well as a defender. Expose your reasoning to people who disagree with you. Keep score. And remember that the feeling of having thought carefully is not the same as having thought fairly.

WHAT THE RESEARCH DOES NOT SAY

- **Intelligence is not useless or harmful.** On average, cognitive ability is associated with many good outcomes, including better performance on many reasoning tasks and somewhat better discernment of true and false news. The point is that it does not reliably protect against motivated reasoning on identity-linked topics.
- **"Smarter people are more biased" is not established.** The finding that numeracy or political knowledge *amplifies* bias has not replicated consistently. The more defensible claim is that greater ability does not *reduce* bias as much as we might hope.
- **Tetlock's research is not proof that experts are worthless.** Experts in his studies often outperformed laypeople and had valuable knowledge. The findings concern long-range forecasting of complex political events, which is unusually difficult. In fields with clear feedback, such as weather forecasting, expertise performs well.
- **We cannot diagnose historical figures.** The interpretation of Doyle's motives offered here is plausible and widely shared by historians, but it remains an interpretation.
- **The 2016 "choice blindness" study** involved specific reasoning problems in a laboratory setting. It shows a striking tendency, not a fixed proportion that applies in everyday life.

Bridge to the next chapter

Several times in this chapter, the solution has pointed towards other people: groups that disagree, critics who check our reasoning, conversations that expose our weak arguments. But other people are not only a remedy. Groups can also build and protect shared illusions that no individual would sustain alone. The next chapter looks at how families, communities, crowds and political tribes create their own versions of reality.

REFLECTION EXERCISE: ARGUING AGAINST YOURSELF

1. **Choose a belief you are proud of having reasoned your way to.** It could be about politics, economics, religion, education or a personal decision.
2. **Write down your three strongest arguments for it.** Take your time and make them good.
3. **Now imagine you are a skilled, fair-minded person who holds the opposite view.** Write that person's three strongest arguments. Not caricatures: arguments that a thoughtful opponent would actually recognise.
4. **Swap the labels.** Imagine that your original three arguments were written by a stranger who holds a view you dislike, and evaluate them as critically as you would evaluate that stranger's reasoning. What weaknesses appear?
5. **Score yourself honestly:** which side did you work harder for? Where did you find it easiest to spot flaws? What does this tell you about how your intelligence was deployed?

KEY TAKEAWAYS

- Intelligence and rationality are **different things**: the capacity to think well is not the same as the disposition to think fairly.
- **Myside bias** appears to be largely unrelated to intelligence; clever people are roughly as prone to it as anyone else.
- Greater knowledge and ability can make us **better at justifying** what we already believe.
- Most people believe they are **less biased than others**, and this blind spot is not reduced by intelligence.
- Expertise held as a fortress (the **hedgehog**) predicts worse than expertise held as a set of revisable hypotheses (the **fox**).
- We judge our own arguments more critically when we **think they belong to someone else**.
- The remedy is to put intelligence **under supervision**: expose reasoning to disagreement, keep score and actively seek disconfirming evidence.

06

CHAPTER 6

Groups, Politics and Shared Illusions

> *Alone, a person may doubt. In a crowd that has already decided, doubt starts to feel like disloyalty.*

The night the sea turned sweet

On the night of Friday, 18 August 2006, a rumour began to spread through the Mahim area of Mumbai. The water of Mahim Creek, near the dargah of a revered fourteenth-century Sufi saint, had turned sweet.

People came to taste it. They told others, who came too. Through the night and into Saturday, thousands gathered at the water's edge. Police estimated about five thousand people were there overnight. Many cupped the water in their hands and drank. Some filled bottles to take home to relatives. Parents gave it to their children. Some attributed the change to the blessing of the saint; oth-

ers simply believed something miraculous had happened and that the water might heal illness.

Mahim Creek is where the Mithi river meets the sea. At the time, it received enormous volumes of untreated sewage and industrial waste every day. Health officials warned that the water was heavily contaminated. Maharashtra's chief minister publicly advised people not to drink it.

The explanation for the "sweetness" was not mysterious. Mumbai had experienced heavy monsoon rains. Reservoirs had overflowed, sending large amounts of fresh water down the Mithi river. Fresh water is lighter than salt water and can form a less salty layer at the surface, especially at low tide in a semi-enclosed creek. By Saturday afternoon, as the tide turned, people at the creek agreed that the water had become salty again.

No single person invented the Mahim miracle. It was assembled collectively, through conversation and observation, by people who each added a little confidence to everyone else's. Some tasted water that was genuinely less salty than usual. Others tasted it after hearing it was sweet and found that, yes, it was. Others did not taste it at all but drank it because so many others were drinking it. The crowd itself became evidence.

This chapter is about how groups create beliefs that their members would not have created alone, and how those beliefs become hard to question from within.

The power of the person next to you

In the early 1950s, the American psychologist Solomon Asch conducted a series of experiments that are among the best known in psychology. Participants sat in a room with several other people and were shown a line, followed by three comparison lines of

clearly different lengths. Their task was to say which comparison line matched the first one. The answer was obvious.

But the other people in the room were not genuine participants. They were working with the experimenter. On certain trials, they all gave the same wrong answer, confidently, one after another. The real participant answered near the end.

Across the critical trials, participants went along with the obviously wrong majority roughly a third of the time. Around three-quarters of participants conformed at least once. Many others held their ground but reported feeling anxious, confused or uncertain. When interviewed afterwards, some conformers said they had doubted their own eyes; others knew the group was wrong but did not want to stand out.

Two kinds of social influence are at work here, and both matter for self-deception:

- **Informational influence**: we use other people's judgements as evidence about reality. When many people believe something, it seems more likely to be true. This is often sensible; we cannot check everything ourselves.
- **Normative influence**: we want to be accepted and to avoid the discomfort of standing alone. We go along to get along.

Over time, the second can quietly become the first. What began as public agreement for the sake of harmony can turn into private belief. The press secretary from the introduction is skilled at finding reasons for whatever the group has already decided.

Conformity also varies with culture. A 1996 meta-analysis of studies using Asch's method, by Rod Bond and Peter Smith, found that conformity was generally higher in more collectivist societies, where harmony with the group is especially valued, than in more individu-

alistic ones. This does not make any culture more gullible. Valuing the group is not a flaw. But it does mean that the social costs of dissent, and therefore the pressures towards shared belief, differ from place to place. In many Indian settings, from family gatherings to workplaces with strong hierarchies, openly contradicting the group, or an elder, can carry real costs.

When everyone privately disagrees

Sometimes a group appears united in a belief that most of its members do not actually hold. Psychologists call this **pluralistic ignorance**.

In a well-known series of studies in the early 1990s, Deborah Prentice and Dale Miller surveyed students at Princeton University about campus drinking. Most students reported that they were personally less comfortable with heavy drinking than they believed the average student was. In other words, many individuals privately had reservations, but each believed that everyone else was relaxed about it. Each student, trying to fit in with what they thought was the norm, behaved as though they agreed, which reinforced everyone else's mistaken impression.

Pluralistic ignorance helps explain why harmful norms can persist long after most people have stopped believing in them. Think of practices such as expensive wedding customs, rigid expectations about career choices or unspoken rules in a workplace. Many participants privately think the practice is excessive. Each assumes others are committed to it. No one wants to be the first to object. The silence of each person becomes evidence for the others that the norm is widely supported.

Why groups drift to the extremes

When like-minded people discuss an issue, they tend to end up holding a more extreme version of the view they started with. This is called **group polarisation**, and it has been observed in many settings since the 1960s.

In one study published in 2007, David Schkade, Cass Sunstein and Reid Hastie brought together small groups of citizens in two American cities. One city, Boulder, was known for liberal politics; the other, Colorado Springs, for conservative politics. The groups discussed contested issues such as climate change and affirmative action. After discussion, the liberal groups had become more liberal and the conservative groups more conservative. Views within each group also became more uniform. Discussion had widened the gap between the cities.

Several mechanisms seem to be at work. In a like-minded group, most of the arguments people hear support the group's leaning, so the balance of arguments tilts further in that direction. People also compare themselves with others and want to be seen as good members of the group, which can mean taking a slightly stronger position than average. And as everyone agrees, confidence grows, and with it the willingness to hold more extreme views.

Anyone who has watched a family or neighbourhood WhatsApp group over a few years may have seen something similar. A group that begins with mild shared opinions about politics, health or local issues can, through a steady stream of forwarded messages and approving replies, become steadily more certain and more extreme, while members who disagree fall silent or leave.

Us and them

The groups we belong to shape not only what we believe but how we see people outside them.

In the early 1970s, the social psychologist Henri Tajfel and his colleagues in Bristol showed how little it takes to create group loyalty. They divided schoolboys into groups on trivial grounds, such as whether they had supposedly preferred paintings by Paul Klee or by Wassily Kandinsky. The boys did not know who else was in their group and would never interact with them. Yet when given the chance to distribute small rewards, they tended to favour members of their own group. Sometimes they even chose options that maximised the *difference* between their group and the other group, rather than the total reward to their own group.

These **minimal group** studies suggest that the mere act of categorising people into "us" and "them" is enough to produce some favouritism. In real life, with long histories, real conflicts and strong emotions, these tendencies can become far more powerful.

The party over the policy

Political loyalty is one of the most powerful forms of group identity, and its effects on reasoning have been studied extensively.

In a striking 2003 study, the psychologist Geoffrey Cohen asked American participants who identified as liberal or conservative to evaluate a welfare policy. Some read about a generous policy; others read about a strict one. Crucially, some were also told which party supported it. When no party information was given, liberals tended to prefer the generous policy and conservatives the strict one, as you might expect. But when told that their own party supported a policy, participants tended to support it, whether it was

generous or strict. Party endorsement had a bigger effect than the content of the policy.

Even more striking, participants did not believe that the party label had influenced them. They said their judgements were based on the details of the policy and their own values. They did, however, believe that *other people* would be swayed by party labels.

Political scientists in many countries have documented growing **affective polarisation**: not just disagreement about policies but dislike and distrust of supporters of the other side. In the United States, studies by Shanto Iyengar, Sean Westwood and others have found that many partisans now express stronger hostility towards the opposing party than they did a few decades ago, and that this hostility can influence judgements in areas that have nothing to do with politics.

India's political landscape is very different from that of the United States, with many parties, regional identities and shifting alliances. Comparable research on Indian voters is still developing, and claims about how polarised India has become should be made carefully. But anyone who follows political debates on Indian social media can recognise some of the same patterns: the tendency to judge a policy by who proposed it, to excuse one's own side's failings as unavoidable and to see the other side's failings as proof of bad character. These patterns appear on every side of the political spectrum. **No party, ideology or community is immune.** If you notice them only in your opponents, that is itself a symptom of the bias blind spot described in Chapter 5.

CASE STUDY: THE MONKEY MAN OF DELHI

In April and May 2001, residents in parts of the Delhi region, starting in Ghaziabad and spreading to East Delhi and Noida, began reporting attacks by a mysterious creature. Descriptions varied widely.

Some said it was a monkey-like being about four feet tall. Others described a man in a mask, a creature with metal claws, a figure that could leap between rooftops or one with glowing lights on its body. It was blamed for scratches, bites and injuries.

The panic spread rapidly. People slept with sticks beside them. Neighbourhood groups organised night patrols. Police offered a reward for the creature's capture and set up special teams. Media coverage, local and international, amplified the reports.

The consequences were real and tragic. At least two people died after falling from buildings or stairs while fleeing what they thought was the creature. Innocent people, including a small, wandering sadhu, were beaten by mobs who mistook them for the Monkey Man.

Then, as suddenly as it had begun, the panic faded. No creature was ever found. A police investigation concluded that the attacks were not real. A study published in 2003 in the *Indian Journal of Medical Sciences* by forensic doctors who examined many of the alleged victims found that the injuries were consistent with accidents, such as falls or scrapes while fleeing, rather than attacks. Most incidents had occurred at night, often during power cuts, in crowded neighbourhoods where many families slept on rooftops during the hot summer.

The Monkey Man case shows how a shared illusion can be built from ordinary ingredients:

- **Ambiguity**: darkness, power cuts and the noises and shadows of a dense city at night.
- **Anxiety**: a hot, uncomfortable summer in crowded conditions, with existing worries about safety.
- **Social proof**: once neighbours and the media reported attacks, each new report seemed to confirm the others.
- **Interpretation**: scratches, falls and frightening experiences were interpreted through the lens of the story.

- **Group action**: patrols and gatherings kept people focused on the threat and made it feel more real.

No individual resident invented the Monkey Man, and most of the people involved were not foolish or dishonest. They were frightened people in a frightening situation, taking cues from each other. That is exactly what makes shared illusions so powerful.

Shared illusions in ordinary life

Most shared illusions are less dramatic than the Monkey Man. They are the comfortable stories that families, companies, institutions and communities tell about themselves.

- A **family** may maintain that a relative's drinking is "just social", because acknowledging the problem would require difficult conversations and change.
- A **company** may convince itself that a failing product is about to turn the corner, because the leadership has publicly committed to it and the employees' jobs depend on it.
- A **school or university** may believe its teaching is excellent because its toppers do well, while overlooking students who struggle.
- A **community** may hold a flattering view of its own history while dismissing uncomfortable parts as exaggerations by outsiders.

In each case, the illusion is maintained not by any individual's self-deception alone but by a network of mutual reassurance. Each member's confidence supports the others'. Questioning the story carries a social cost. Over time, the story becomes simply "how things are".

The psychologist Irving Janis coined the term **groupthink** in the early 1970s to describe how cohesive decision-making groups, such

as a president's advisers planning a military operation, could suppress doubts and reach disastrous decisions. The term has entered everyday language, and the warning it captures is important. As the box below notes, however, the specific model has received mixed support in later research.

WHAT THE RESEARCH DOES NOT SAY

- **Asch did not show that people are sheep.** Even in his studies, most responses on critical trials were independent, and about a quarter of participants never conformed. Later researchers have emphasised that the studies also show how often people resist group pressure.
- **Conformity differences between cultures are averages.** They say nothing about any individual and do not mean that collectivist cultures produce less independent thinkers.
- **"Groupthink" is a useful warning, not a well-validated theory.** Studies testing Janis's full model have produced mixed results. Cohesive groups do not always make worse decisions; sometimes cohesion helps.
- **Robbers Cave and the Stanford Prison Experiment are not reliable foundations.** These famous studies of group conflict and roles are often cited in popular books. Later investigations into their methods have raised serious concerns, including the extent to which researchers influenced the outcomes. This book does not rely on them.
- **"Filter bubbles" and "echo chambers" are often overstated.** Several large studies have found that most people's media diets are more mixed than the popular image suggests, and that a smaller group of highly engaged partisans accounts for much of the one-sided consumption. Online groups matter, but they are not sealed chambers.
- **Research on Indian political polarisation is still developing.** Patterns documented in the United States should not be assumed to apply in the same way in India.

Bridge to the next chapter

Groups do not only create shared beliefs internally. They also spread them outward. A rumour that begins in one neighbourhood

can, in the age of smartphones, reach millions within hours. The next chapter examines why false stories travel so well, why they are so hard to correct, and why, in some tragic cases, they have led to violence.

REFLECTION EXERCISE: AUDITING YOUR GROUPS

1. **List three groups whose opinions strongly influence yours.** These could be your family, a WhatsApp group, a set of friends, a professional community, a political or religious community or a set of accounts you follow online.
2. **For each group, write down one belief that almost everyone in that group shares.** Then ask: how did I come to hold this belief? Did I examine the evidence, or did I absorb it from the group?
3. **Think about the last time someone in one of these groups disagreed with the majority.** What happened to them? How did you respond? Would you be willing to be that person?
4. **Look for possible pluralistic ignorance.** Is there a norm in one of your groups that you privately doubt? Is it possible that others doubt it too?
5. **Try a small experiment:** in a low-stakes way, express a mild private doubt in one of these groups, with curiosity rather than accusation. Notice how it feels and what happens.

KEY TAKEAWAYS

- We are influenced by groups both because we use others' views as **evidence** and because we want to **belong**.
- Many people conform to obviously wrong majority views at least occasionally, and the strength of conformity **varies across cultures**.
- **Pluralistic ignorance** can keep norms alive even when most members privately doubt them.
- Like-minded groups tend to **drift towards more extreme** positions through discussion.
- Even trivial group divisions can produce **favouritism**, and political loyalty can shape how we judge policies more than their content does.
- Shared illusions are built from **ambiguity, anxiety, social proof and collective interpretation**, as the Mahim and Monkey Man episodes show.
- No group, party or community is immune; noticing these patterns only in opponents is itself a warning sign.

07

CHAPTER 7

How False Stories Spread

> *A false story does not need to be believed by everyone. It only needs to be believed by enough people, at the right moment, in the right place.*

Sunday market, Rainpada

On Sunday, 1 July 2018, a weekly market was under way in Rainpada, a tribal village in Maharashtra's Dhule district. A state transport bus arrived, and a group of men got off. They belonged to the Nath Panthi Davari Gosavi, a nomadic community whose members travel from place to place, often seeking alms. One of them was seen speaking to a young girl.

For days, messages had been circulating in the area, on WhatsApp and by word of mouth, warning that gangs of child kidnappers were active. Some versions claimed the kidnappers took children's organs. People in the market grew suspicious of the strangers. A crowd formed. Five of the men were dragged into the village panchayat office and beaten to death.

The victims were not kidnappers. They were travellers who had come, like many before them, to a market. The police arrested more than twenty people in the following days. The Dhule superintendent of police told reporters that the accused came from a village where messages about child kidnappers had gone viral.

Rainpada was not an isolated event. In the months around it, false rumours of child kidnappers, spread largely through WhatsApp, were linked to mob attacks in many states across India, from Tripura and Assam to Tamil Nadu, Telangana, Karnataka and Gujarat. Fact-checkers counted more than twenty deaths in the first half of 2018 alone. In one bitterly ironic case in Tripura, a man employed by the state government to go from village to village warning people not to believe such rumours was himself killed by a mob that suspected him of being a kidnapper.

In almost all of these places, there was no evidence of an actual wave of child abductions. The danger was invented. The deaths were real.

How does a false story become powerful enough to kill?

The arithmetic of a rumour

In 1947, as the Second World War receded, the American psychologists Gordon Allport and Leo Postman published *The Psychology of Rumor*. They had studied wartime rumours and proposed a simple formula. The strength of a rumour, they suggested, depends on two things multiplied together: **how important** the subject is to the people hearing it, and **how ambiguous** the available information is.

If a topic does not matter to us, a rumour about it will not spread far, however uncertain the facts. If the facts are clear and widely known, a rumour has little room to grow, however important the

topic. But when something matters intensely and clear information is scarce, rumours flourish.

The child-kidnapping rumours of 2018 fitted this formula exactly. Few things matter more to parents than the safety of their children. And in many of the affected areas, reliable local information was hard to come by: police statements might not reach villages quickly, local news coverage was limited, and people had little way of checking whether a message from a distant city was true. Strangers, especially those who looked or spoke differently, became the focus of fear.

Allport and Postman's formula was intended as a rough guide rather than a precise equation, and later researchers have added other factors, including anxiety, uncertainty and how credible the rumour seems. But the core insight remains useful. **Rumours fill the space between what matters and what is known.**

Faster, further, deeper

In 2018, a team at the Massachusetts Institute of Technology, led by Soroush Vosoughi, Deb Roy and Sinan Aral, published one of the largest studies ever conducted of how true and false news spreads online. They examined around 126,000 stories that had spread on Twitter between 2006 and 2017, shared by about three million people, and classified them as true or false using the judgements of six independent fact-checking organisations.

The findings were stark. False stories spread farther, faster, deeper and more broadly than true ones, in all categories of information. The effect was strongest for political news. False stories were about seventy per cent more likely to be shared than true ones. True stories took around six times as long as false ones to reach 1,500 people.

The researchers also checked whether automated accounts, or "bots", were responsible. They found that bots spread true and false news at about the same rate. The difference came from **human beings**. People were the ones choosing to share falsehood more.

Why? The researchers found that false stories tended to be more **novel** than true ones: they contained information people had not encountered before. And the replies to false stories expressed more **surprise and disgust**, while replies to true stories expressed more sadness, anticipation and trust. Novelty and strong emotion seem to make information more shareable. False stories, freed from the constraints of reality, can be engineered to be more novel and more emotional than the truth usually is.

Other research supports the role of emotion. In a 2017 study of messages on Twitter about contested political issues in the United States, William Brady and colleagues found that each additional moral-emotional word in a message, words like *shame*, *evil* or *attack*, was associated with a substantial increase in how often it was shared, mainly within the sender's own political group.

Why repetition becomes truth

Once a false claim is circulating, another psychological effect helps it settle in. In 1977, Lynn Hasher, David Goldstein and Thomas Toppino asked participants to rate how true various trivia statements were, over several sessions. Some statements were repeated across sessions; others were new. Repeated statements were rated as more likely to be true, whether they were actually true or false.

This is now known as the **illusory truth effect**, and it is one of the most reliable findings in the psychology of belief. The likely explanation is that repeated information is easier to process. It feels familiar, and the mind treats that ease, or *fluency*, as a sign of truth. A

claim we have heard before feels right in a way that a new claim does not.

Perhaps most worrying, a 2015 study led by Lisa Fazio found that the effect occurred even when participants had relevant knowledge that contradicted the claim. People who knew that the short pleated skirt worn by Scots is called a kilt were still somewhat more likely to rate as true the false statement that it is called a sari, if they had seen that statement before. Knowledge helps, but it does not make us immune to repetition.

In a world of forwarded messages, repetition is easy to achieve. The same claim arrives from a cousin, a colleague, a neighbourhood group and a family group. Each time, it feels a little more familiar, and therefore a little more true. And because it comes from people we know and trust, we may lower our guard further.

CASE STUDY: THE CHIP IN THE NOTE AND THE SALT THAT NEVER RAN OUT

On the evening of 8 November 2016, Prime Minister Narendra Modi announced that ₹500 and ₹1,000 notes would cease to be legal tender, and that new ₹500 and ₹2,000 notes would be issued. It was one of the most dramatic economic announcements in India's history, affecting almost every household. It was also, for millions of people, deeply confusing. That combination of importance and ambiguity, as Allport and Postman would have predicted, created ideal conditions for rumours.

Within hours, a message began circulating on WhatsApp and other platforms claiming that the new ₹2,000 note contained a "nano GPS chip" that allowed authorities to track it by satellite, even, according to some versions, when buried deep underground. The idea was presented as a clever way to catch black money.

The claim was false. The Reserve Bank of India's spokesperson told reporters that no such technology existed. The finance minister also

dismissed it. Science writers pointed out the obvious problems: a chip capable of communicating with satellites would need a power source and an antenna, and would cost far more than the note itself to produce.

Yet the rumour spread widely, and some versions were repeated on television before being corrected. Why did it find so many believers?

- **It was novel and surprising.** A satellite-tracked banknote sounded like something out of a science-fiction film.
- **It fitted a story people wanted to believe.** For those who supported the demonetisation policy, the chip offered a satisfying explanation of how it would catch wrongdoers. For others, it fitted fears of surveillance. A story that serves more than one identity can spread across groups.
- **It was hard for most people to check.** Few had the technical knowledge to evaluate the claim, and it arrived through trusted contacts.

A few days later, a different rumour spread: that salt was running out and prices were soaring because of the cash shortage. In several cities, people rushed to shops, sometimes late at night, and bought salt in bulk. Prices rose sharply in some places. Police and government officials issued statements denying any shortage. India, with its long coastline, produces far more salt than it needs. The panic subsided within days.

The salt rumour illustrates another mechanism: the **self-fulfilling rumour**. Once people believed a shortage was coming, their panic buying created a temporary local shortage, which then seemed to confirm the rumour. The story generated its own evidence.

Neither rumour caused the kind of tragedy seen at Rainpada. But together they show how quickly false stories can emerge when something important is uncertain, and how they attach themselves to whatever people already fear or hope.

Why corrections don't fully undo the damage

Once a false claim has been believed, correcting it is harder than one might hope. Not impossible, but harder.

Researchers call this the **continued influence effect**. In a classic paradigm developed in the 1990s by Hollyn Johnson and Colleen Seifert, participants read a series of reports about a warehouse fire. Early reports suggested that the fire was caused by volatile materials, paint and gas cylinders, stored carelessly in a cupboard. A later report corrected this: the cupboard had in fact been empty. When participants were later asked questions about the fire, many continued to rely on the volatile materials explanation, even though they remembered the correction.

Why? One explanation is that the false information filled a gap in the participants' mental model of the event. The correction removed it but left a hole: if not the paint, then what? In the absence of an alternative explanation, people fell back on the one they had. When researchers provided a plausible alternative cause, such as evidence of arson, the continued influence of the original misinformation was substantially reduced.

This has important implications. **A good correction does not just say "that is false." It also says what is true, and explains why the false claim seemed plausible.** Fact-checkers sometimes call this a "truth sandwich": lead with the fact, briefly mention the falsehood, then return to the fact.

There is good news here too. For years, it was widely believed that corrections often "backfire", causing people to cling even more tightly to false beliefs. As the box at the end of this chapter explains, large-scale research has found that this is rare. In most studies, corrections move people towards the truth. They do not always change attitudes or votes, and they rarely eliminate the mis-

information's influence completely. But they work better than their reputation.

Sharing without thinking

If false stories spread mainly because people share them, why do people share things that are false? It is natural to assume that people share what they believe, and that they believe falsehoods because they are biased. Some recent research suggests an important additional factor: **inattention**.

In a series of studies, the psychologists Gordon Pennycook and David Rand found that people who scored higher on tests of analytical thinking were somewhat better at telling true headlines from false ones, regardless of whether the headlines favoured their political side. They summarised their 2019 finding with a memorable phrase: people who fall for fake news may be "lazy, not biased". In other words, much susceptibility to misinformation may come not from motivated reasoning but from not stopping to think.

They also found a curious gap between what people believe and what they share. When asked whether a headline was accurate, people were reasonably good at distinguishing true from false. But when asked whether they would *share* the same headlines, they were much less discerning. The social media environment seems to focus attention on other things: whether a post is interesting, funny, outrageous or likely to get a reaction from friends. Accuracy slips out of focus.

Building on this, Pennycook, Rand and colleagues tested a simple intervention. Before people browsed headlines, they were asked to rate the accuracy of a single, unrelated headline. This small nudge, which shifted attention towards accuracy, modestly improved the quality of the news people said they would share. Large studies

spanning many countries have since tested similar prompts, with modest but generally positive results.

This is not the whole story. Motivated reasoning, as earlier chapters showed, is real. And researchers continue to debate how much of the spread of misinformation is due to inattention, how much to bias and how much to people sharing things they know may be false because the content serves a social purpose: expressing outrage, signalling loyalty or simply entertaining friends. But the inattention research offers a hopeful message. **Many people do care about accuracy. They just need reminding at the right moment.**

What platforms and policies changed

After the wave of lynchings in 2018, the Indian government pressed WhatsApp to act. In July 2018, the company announced that it would test a limit in India allowing users to forward a message to only five chats at once, and remove the quick-forward button next to photos and videos. WhatsApp said that people in India forwarded more messages, photos and videos than people in any other country. In January 2019, after a test period that the company said significantly reduced forwarding, the five-chat limit was extended to users worldwide. Messages that have been forwarded many times now carry a label, and further restrictions on such messages were introduced later.

Such measures introduce **friction**: small obstacles that slow down impulsive sharing. Friction does not stop determined people, and a message forwarded five times by each recipient can still reach large numbers of people. But by making mass forwarding slightly harder, it reduces the speed at which rumours spread, and gives corrections a better chance to catch up.

Ultimately, though, technology alone cannot solve a problem rooted in human psychology. The deeper solutions involve how individuals think before they share, how communities respond to rumours and how people learn to recognise manipulation. Those are the subjects of the final chapter.

WHAT THE RESEARCH DOES NOT SAY

- **The "backfire effect" is rare.** A famous 2010 study suggested that correcting false political beliefs could sometimes make them stronger. But in 2019, Thomas Wood and Ethan Porter tested 52 contested issues with more than 10,000 participants and found no backfire for any of them. Most later research agrees that corrections usually move people towards accuracy, even on politically sensitive topics.
- **The Vosoughi study is about Twitter between 2006 and 2017.** It is a powerful study, but platforms, algorithms and user behaviour differ. Its findings should not be assumed to transfer exactly to WhatsApp, where messages travel privately and cannot be studied in the same way.
- **"Lazy, not biased" is not the final word.** Inattention explains some susceptibility to misinformation, but motivated reasoning, identity and deliberate sharing for social reasons also play roles. Researchers continue to debate their relative importance.
- **Accuracy nudges have small effects.** They are cheap and can be applied at scale, which makes them worth using, but they are not a cure.
- **WhatsApp alone did not cause the lynchings.** Rumours of child kidnappers have caused mob violence in many times and places without smartphones. Pre-existing fears, prejudice against outsiders, weak trust in institutions and slow police response all mattered. Technology changed the speed and scale; the underlying psychology is old.
- **Allport and Postman's formula is a heuristic**, not a precise law.

Bridge to the next chapter

We have now travelled the full path from the individual mind to the crowd: the need to be right, the defence of identity, the filtering of evidence, the reconstruction of memory, the misuse of intelligence, the pull of the group and the spread of false stories. The obvious question remains: what can we do about it? Not about other people's minds, but about our own. The final chapter turns to the difficult, rewarding art of changing your mind.

REFLECTION EXERCISE: THE PAUSE BEFORE YOU FORWARD

For the next seven days, try this whenever you are about to forward a message, share a post or repeat a claim you heard.

1. **Pause and rate it.** Before sharing, ask yourself: on a scale of one to five, how confident am I that this is true? Write the number down, even mentally.
2. **Name the emotion.** What does the message make you feel: anger, fear, surprise, pride, disgust? Strong emotion is not proof of falsehood, but it is a signal to slow down.
3. **Check the source.** Who originally created this? Is there a named, accountable source? Has a reliable outlet or fact-checker reported it?
4. **Ask the cost question.** If this turns out to be false, who could be harmed?
5. **Keep a tally.** At the end of the week, count how many messages you decided not to forward, and how many you checked and found to be false or misleading. Reflect on what you learned about your own sharing habits.

KEY TAKEAWAYS

- Rumours flourish where a topic is **important** and reliable information is **ambiguous**.
- Large-scale research found that false news on Twitter spread **farther and faster** than true news, mainly because **humans** chose to share it.
- **Novelty and strong emotion** make information more shareable, which gives invented stories an advantage.
- **Repetition makes claims feel true**, even when we have knowledge that contradicts them.
- Corrections work best when they **replace** a false explanation with a true one, rather than simply denying it.
- The feared "backfire effect" is **rare**; most corrections move people towards the truth.
- Much misinformation is shared through **inattention**, and simple prompts to think about accuracy can help.
- **Friction**, such as forwarding limits, slows rumours down and gives truth time to catch up.

08

CHAPTER 8

The Difficult Art of Changing Your Mind

> *Changing your mind is not a defeat. It is the only evidence that you were using it.*

The study everyone wanted to be true

In December 2014, the prestigious journal *Science* published a study that delighted many readers. A graduate student named Michael LaCour, working with Donald Green, a respected political scientist at Columbia University, reported that a single conversation of about twenty minutes with a gay canvasser could durably change Californian voters' views on same-sex marriage. The effect appeared to last for months and even to spread to other members of the voters' households.

The finding was remarkable. Decades of research had suggested that attitudes on contested social issues were extremely hard to shift, especially with brief conversations. Here was evidence that

genuine human contact could do what advertising and argument could not. The study was widely covered in the press. It seemed to confirm something many people hoped was true about the power of empathy.

Two young researchers at the University of California, Berkeley, David Broockman and Joshua Kalla, were among those inspired. They wanted to build on the study, using similar methods to test conversations on a different issue. As they began to plan, they tried to understand exactly how LaCour had conducted his survey, and things started not to add up. The response rates were implausibly high. The survey company that had supposedly run the study said it had never done so. The data looked suspiciously tidy, as though they might have been generated from an existing data set.

In May 2015, Broockman, Kalla and a colleague, Peter Aronow, released a report detailing these irregularities. Donald Green, confronted with the evidence, did something that deserves to be remembered as clearly as the scandal: he promptly asked *Science* to retract the paper, his own paper, stating that he had lost confidence in the data. The journal retracted it later that month.

The story might have ended there, as a cautionary tale. But Broockman and Kalla still wanted to know whether conversations could change minds. So they ran their own carefully designed experiment. Working with a Los Angeles LGBT organisation that had developed a technique called **deep canvassing**, they sent canvassers to homes in Miami to talk with voters about the rights of transgender people. The conversations lasted about ten minutes. Canvassers did not lecture. They asked voters to recall a time when they themselves had been judged negatively for being different, and to reflect on how that might relate to the experiences of transgender people.

The results, published in *Science* in 2016, showed that these brief, non-confrontational conversations reduced prejudice against transgender people, and that the effects were still measurable three months later. It was a smaller effect than LaCour had claimed. It was also real.

This story contains almost every theme of this chapter in miniature. A false finding was believed partly because people wanted it to be true. It was exposed by researchers who looked closely at evidence they had every reason to accept. A senior scientist chose to admit error publicly instead of defending his reputation. And the underlying hope, that minds can change through respectful conversation, turned out to have some truth after all, once it was tested honestly.

Why changing your mind is hard

By now, the obstacles should be familiar. Over the previous chapters we have seen that:

- admitting a mistake threatens our sense of being **competent, moral and sensible** (Chapter 1);
- many beliefs are tied to **identity and belonging** (Chapter 2);
- we **filter evidence**, seeking and accepting what suits us (Chapter 3);
- our **memories** are reconstructed in flattering ways (Chapter 4);
- our **intelligence** can be recruited to defend rather than test our beliefs (Chapter 5);
- **groups** reinforce shared beliefs and punish dissent (Chapter 6);
- and **repetition and emotion** make false stories feel true (Chapter 7).

Given all this, it might seem surprising that anyone ever changes their mind. Yet people do, constantly. Most of us hold different views now from those we held ten years ago. We change our minds about food, health, careers, people and politics. Some of these changes are dramatic; most are gradual.

The question is not whether minds can change. It is **how to make it more likely that our minds change when the evidence calls for it**, rather than only when social pressure, convenience or chance push us. The rest of this chapter offers practical approaches, each grounded in research, for doing that: first with ourselves, then with others.

Part one: changing your own mind

Separate who you are from what you think

The single most important step may be to loosen the link between identity and belief. If a belief is part of who you are, giving it up feels like losing part of yourself. If your identity is instead tied to *how* you think, to being curious, fair and willing to update, then changing a belief can feel like an expression of who you are rather than a betrayal of it.

The writer Julia Galef describes this as the difference between a **soldier mindset** and a **scout mindset**. A soldier's job is to defend territory. A scout's job is to find out what is really there, and to report it accurately even if the news is unwelcome. Both are necessary roles, but when it comes to beliefs, the scout's role is the one that leads to accuracy. Galef suggests that people can deliberately cultivate the scout's identity: taking pride in noticing when they are wrong, rather than shame.

Research on **intellectual humility**, the recognition that one's beliefs might be wrong, offers some support. In studies led by Mark Leary

and others, people who scored higher on measures of intellectual humility were better at distinguishing strong arguments from weak ones, and were less likely to judge people negatively for changing their minds. Intellectual humility is not the same as low confidence or lack of conviction. It is the ability to hold beliefs firmly enough to act on them, and lightly enough to revise them.

India's own intellectual traditions offer a helpful image here. The Jain principle of *anekantavada*, often translated as "many-sidedness" or "non-absolutism", holds that reality is complex and that any single perspective captures only part of the truth. It is often illustrated by the parable, shared across Jain, Buddhist and Hindu sources, of several blind men touching different parts of an elephant: one feels the trunk and says it is like a snake, another the leg and says it is a pillar, another the ear and says it is a fan. Each is partly right. Each is wrong to think he has the whole. The parable has been told for centuries because it captures something modern psychology has rediscovered: **the confidence of a partial view is not evidence of its completeness.**

Write down your predictions and keep score

In Chapter 4, we saw how memory and hindsight bias rewrite our past beliefs, so that we seem to have been right all along. One of the most effective defences is simple: write down what you think *before* you know the outcome.

Philip Tetlock's research with superforecasters, introduced in Chapter 5, found that the best forecasters made specific predictions with numerical probabilities, then checked them against what happened. A prediction like "the economy will struggle next year" cannot be scored. A prediction like "I think there is a seventy per cent chance that inflation will be above five per cent in a year's time" can. Over time, scoring shows you whether you are well cal-

ibrated: whether the things you rate at seventy per cent actually happen about seventy per cent of the time.

You do not need to forecast global events to benefit. You can predict how a project at work will go, whether a new habit will stick, how a family discussion will unfold or whether a share you bought will rise. The act of writing down a prediction creates a record that your memory cannot quietly revise.

Follow Darwin's golden rule

Charles Darwin wrote in his autobiography that he had followed a golden rule for many years. Whenever he came across a published fact, a new observation or a thought that contradicted his general conclusions, he made a written note of it at once. He had found, he explained, that such unwelcome facts were far more likely to slip from memory than favourable ones.

Darwin had discovered, without the language of modern psychology, the asymmetry of confirmation bias described in Chapter 3. His remedy was to counteract it deliberately: to give disconfirming evidence extra weight in his records precisely because his mind would otherwise give it less.

You can adopt the same rule. Keep a note, physical or digital, of evidence that contradicts your important beliefs. You do not have to act on it immediately. You simply have to stop your mind from deleting it.

Consider the opposite

In 1984, Charles Lord, Mark Lepper and Elizabeth Preston returned to the biased evaluation of evidence they and their colleagues had studied in 1979. This time, they tested ways to reduce the bias. Telling people to be "fair and unbiased" had little effect. People already believed they were being fair. But a different instruction

worked better: asking people to consider whether they would have evaluated the study the same way if it had produced the *opposite* result.

This "**consider the opposite**" strategy works because it directly targets the asymmetry in our standards. If I find a study's methods convincing when it supports my view, I should ask whether I would find the identical methods convincing if the study had reached the other conclusion. If not, my evaluation is being driven by the result, not the method.

Make it safe to be wrong

In fields where mistakes can be catastrophic, such as aviation and surgery, organisations have learned that punishing error does not eliminate it. It drives it underground. People hide mistakes, which means they cannot be learned from. Many airlines and hospitals now use reporting systems in which errors and near misses can be disclosed with little fear of punishment, so that problems are found and fixed before they cause harm.

The same principle applies to individuals and families. If admitting a mistake brings ridicule, people learn not to admit mistakes, including to themselves. Parents who respond to a child's confession with anger may teach the child to hide errors. Managers who punish bad news teach teams to report only good news. And people who mock others for changing their minds make it harder for anyone, including themselves, to change.

You can start by changing the way you talk about your own mistakes. Saying "I was wrong about that, and here is what I learned" in front of others signals that error is survivable. It also builds the identity of a scout.

Part two: helping others think more clearly

Most of us are interested in changing our own minds in theory. We are much more interested, in practice, in changing other people's. The research suggests some uncomfortable truths about this, and some useful approaches.

Do not humiliate

The first rule is the simplest and most often broken. If changing one's mind means admitting that one is foolish, people will resist. Mockery, sarcasm and public "gotcha" moments can feel satisfying, and they may impress people who already agree with you. They rarely persuade the person being mocked. As Chapter 2 showed, people become more open when changing their minds does not require them to lose face.

Ask, don't tell

Deep canvassing, the technique tested by Broockman and Kalla, relies heavily on questions rather than arguments. Canvassers asked people about their own experiences and values, listened carefully and invited them to reflect. A later study by Kalla and Broockman, published in 2020, suggested that what mattered most was a **non-judgemental exchange of personal stories**: listening to the voter's experiences and sharing relevant experiences in return, rather than simply presenting facts or arguments.

A related approach, sometimes called **street epistemology**, involves asking people how confident they are in a belief and *how they came to know it*, rather than challenging the belief directly. Questions like "What would change your mind?" or "How would you know if you were wrong?" invite reflection without attack. Their aim is not to win an argument, but to help the other person examine their own reasoning.

Lead with what you share

People are more open to information from those they see as part of their own group. A message that comes from a trusted member of a community often lands very differently from the identical message delivered by an outsider seen as hostile. If you want to discuss a contentious issue with a family member, beginning with values you genuinely share, care for the grandmother's health, say, in the family argument from Chapter 2, can reduce the sense of threat.

Replace, don't just remove

Chapter 7 showed that corrections are more effective when they provide an alternative explanation. If you tell someone that a rumour is false, try also to explain what is actually going on, and why the rumour seemed believable. Simply removing a belief leaves a gap that the mind may fill with the old story.

Prebunk instead of debunk

The most effective time to address misinformation may be before people encounter it. In the 1960s, the social psychologist William McGuire proposed what he called **inoculation theory**. Just as a vaccine exposes the body to a weakened form of a virus, he suggested, people could be "inoculated" against persuasion by being exposed to weakened versions of misleading arguments, along with explanations of why they are wrong.

In recent years, researchers including Sander van der Linden and Jon Roozenbeek at the University of Cambridge have developed this idea into practical tools. Rather than refuting specific false claims one by one, they teach people to recognise common **manipulation techniques**: emotionally manipulative language, false dichotomies ("either you support this or you hate the country"), scapegoating, fake experts and conspiratorial framing. In one large study published in 2022, short animated videos explaining such techniques,

shown as advertisements on YouTube, improved viewers' ability to recognise them.

This approach has been taken to India. In December 2022, Jigsaw, a unit of Google, announced a prebunking campaign in India, working with local partners to produce short videos in Bengali, Hindi and Marathi. The campaign was explicitly framed as an opportunity to test prebunking outside Western countries, in one of the world's largest and most linguistically diverse information environments.

CASE STUDY: TEACHING CHILDREN TO DOUBT IN KANNUR

In October 2017, the district administration of Kannur in northern Kerala was running a vaccination campaign against measles and rubella. Uptake was disappointing. Parents of many children refused to let them be vaccinated, influenced by claims circulating on social media that the vaccine was harmful.

The district administration issued a directive that parents who did not want their children vaccinated would have to obtain the collector's signature. To the collector's surprise, people began queuing outside his office to seek exemptions. According to an account in *The News Minute*, when Collector Mir Mohammed Ali asked one parent why he objected, the reply was that vaccination was harmful, he had seen it on WhatsApp.

The experience convinced Ali that misinformation could not be tackled only with official denials. In June 2018, the Kannur district administration launched an initiative called *Satyameva Jayate*, "Truth alone triumphs". Teachers were trained to give classes to students in around 150 government schools, mainly in classes 8 to 12, on what fake news is, why it spreads and how to recognise it. Students were shown examples of false messages and asked what they would do if they received them. The collector argued that schoolchildren were important targets, not only for their own sake but because many of them had access to their parents' phones and

could influence how their families responded to forwarded messages.

The initiative received national and international attention, including coverage by the BBC, and Kerala later expanded digital literacy efforts more widely. Its long-term impact has not been rigorously measured in the way that a scientific trial would require, so it should not be presented as proof of a technique. But it illustrates several principles of this chapter:

- **Starting before people are exposed.** Teaching students to recognise false messages is a form of prebunking.
- **Building a habit of questioning**, not just correcting individual claims.
- **Using trusted messengers.** Teachers, and through them students, can reach families in ways that official statements cannot.
- **Shared responsibility.** As the collector put it in interviews, everyone tends to believe they play no role in spreading fake news, when in fact many of us do.

Be patient

Finally, minds change slowly. People rarely announce on the spot that they were wrong. More often, a conversation plants a seed. Weeks or months later, the person may quietly adjust their view, often without remembering, as Chapter 4 would predict, that they ever thought differently. If you measure success by whether someone concedes immediately, you will usually be disappointed. If you measure it by whether you have left them with something to think about, and a relationship in which they can think about it without losing face, you will often do better.

The point of all this

It would be possible to read this book and come away cynical: humans are irrational, beliefs are tribal badges and truth is hopeless. That is not the conclusion the evidence supports.

The same research that documents our biases also shows that people respond to good evidence more often than not, that corrections usually work, that conversation can change minds, that groups can correct individual errors, and that habits of careful thinking can be learned. The mind that lies to itself is also the mind that can catch itself in the act. Darwin did it with a notebook. Donald Green did it by retracting his own paper. Jennifer Thompson did it by facing the man she had wrongly identified and working with him to reform the system that had failed them both. Ordinary people do it every day, in small ways that no one notices.

The goal is not to become a perfectly rational being. No such being exists. It is to become a slightly better scout: a little more curious about where you might be wrong, a little more generous towards people who are wrong in different ways, and a little more willing to say the three hardest words in any language: *I was wrong.*

WHAT THE RESEARCH DOES NOT SAY

- **Deep canvassing is not a magic technique.** Effects in Broockman and Kalla's studies were real but modest, and later studies on other issues have found varying results. It works better on some topics, with some people, than others.
- **Prebunking effects fade.** Inoculation studies show measurable improvements in recognising manipulation, but the effects tend to decay over time unless reinforced with "booster" exposure. Effects on real-world sharing behaviour are harder to measure than effects in surveys.
- **Intellectual humility research is still developing.** Measures vary, and much of the evidence is correlational: it shows that humble people behave differently, not necessarily that training humility causes better reasoning.
- **The Kannur initiative has not been formally evaluated.** It is offered as an illustrative example of an approach that is consistent with research, not as evidence of effectiveness.
- **Not every belief deserves equal doubt.** Intellectual humility does not mean treating all claims as equally valid. Some claims, such as that vaccines approved after rigorous testing cause more harm than good, have been tested extensively and found wanting. Being open-minded means following evidence, not abandoning it.

REFLECTION EXERCISE: YOUR PERSONAL UPDATE LOG

This final exercise is designed to become a habit rather than a one-off task.

1. **Start an "update log".** Use a notebook or a note on your phone. Title it something like *Things I have changed my mind about.*
2. **Fill in the first entries from your past.** List at least three beliefs you held five or ten years ago that you no longer hold. For each, note what changed your mind: evidence, experience, a person, a conversation.
3. **Add a "Darwin page".** Whenever you encounter evidence that contradicts one of your important beliefs, write it down immediately, even if you are not yet persuaded.
4. **Add a "predictions page".** Write down two or three specific predictions, with a probability and a date by which you will check them.
5. **Review the log monthly.** Each time you update a belief, however slightly, record it. Over time, the log becomes evidence, for you, that changing your mind is something you do, and something you can be proud of.

KEY TAKEAWAYS

- Minds change all the time; the goal is to make them change **in response to evidence** rather than only to pressure or convenience.
- Loosen the link between **identity and belief**: take pride in how you think, not just in what you think.
- **Write down predictions** and keep score, so that memory cannot rewrite your past.
- Follow **Darwin's golden rule**: record contradicting evidence immediately, because your mind will otherwise forget it.
- Use **"consider the opposite"**: would you judge the evidence the same way if it pointed the other way?
- Make it **safe to be wrong**, for yourself and others.
- When talking to others: **don't humiliate, ask rather than tell, lead with shared values, replace false explanations and be patient**.
- **Prebunking**, teaching people to recognise manipulation before they encounter it, is a promising and scalable defence.

Conclusion: Living Honestly with a Mind That Lies

We began with a particular kind of lie: the one with only one mind in it, where the teller and the listener are the same person and the gap between them has closed.

Over eight chapters, we have watched that lie at work. We saw a group in Chicago reinterpret a failed prophecy as a triumph. We saw a celebrated chairman ride a tiger he could not dismount. We saw doctors resist the evidence that their own hands were carrying death, and thousands of devotees watch milk run down the face of an idol without seeing it run. We saw a woman who studied her attacker's face with all her strength and still chose the wrong man, and a brilliant writer defend paper fairies against the testimony of the girls who had cut them out. We saw a city drink from a sewage-laden creek, a capital hunt for a creature that never existed and a village market turn into a scene of murder because of a message no one had verified.

It would be easy to leave these stories with a sense of superiority, or of despair. Both responses miss the point.

Not a flaw in other people

The first lesson is that **these are not stories about unusual people.** The machinery that produced them is ordinary. The need to protect our sense of being competent and good, the tendency to treat

beliefs as badges, the habit of searching for confirmation, the reconstruction of memory, the recruitment of intelligence to defend conclusions, the pull of the group and the appeal of novel, emotional stories: these are features of human minds in general. They operate in the professor and the farmer, the doctor and the devotee, the rationalist and the believer, the reader of this book and its author.

That should make us more humble, and also more kind. When we see someone holding a belief that seems absurd to us, the most accurate explanation is rarely that they are stupid or wicked. More often, they are a person like us, whose standard-issue mind has been applied to a particular set of experiences, loyalties, fears and information. If we had their history and their circumstances, we might believe what they believe.

Not a counsel of despair

The second lesson is that **the situation is far from hopeless.** The research that reveals our biases also reveals our capacities for correction. People update their beliefs in response to clear evidence more often than popular accounts suggest. Corrections usually help. Groups that genuinely disagree can catch errors that individuals miss. Habits like writing down predictions, recording contradicting evidence and considering the opposite measurably improve judgement. Respectful conversation can shift even deeply held attitudes. Communities can teach their children to question what they are told.

None of these remedies is perfect, and none works every time. But together they suggest that the mind's tendency to deceive itself is not destiny. It is a default, and defaults can be overridden with practice.

Three habits to keep

If you take only three habits from this book, let them be these.

First, notice the pull. When you feel a strong desire for something to be true, or a strong resistance to something being true, treat that feeling as a signal. It does not mean you are wrong. It means your press secretary is likely to be active, and that your evaluation of the evidence deserves a second look.

Second, keep records. Your memory will flatter you; your notes will not. Write down predictions before outcomes are known. Write down evidence that contradicts your views. Write down the things you have changed your mind about. Over time, these records become a more honest account of your thinking than your memory can provide.

Third, make room for being wrong, in yourself and in others. Treat changing your mind as a sign of strength. Speak about your past mistakes without shame. When others change their minds, welcome them rather than mocking them for having been wrong. Every time you make it easier for someone, including yourself, to say "I was wrong", you make the world slightly more honest.

The last word

A central idea in this book is that self-deception thrives in the space between what we want to be true and what is true. That space never closes completely. We are creatures with desires and loyalties, and we always will be.

But we can learn to see the space. We can learn to stand in it with curiosity rather than fear. We can learn to ask, as often as we can bear, a simple question: *What would I see if I were not so invested in seeing something else?*

That question will not make you infallible. It will make you a better witness to your own mind. And in a world full of confident voices, that may be one of the most valuable things a person can be.

The Self-Honesty Toolkit

A short summary of the practical techniques in this book. Keep it somewhere you will see it.

When you feel very sure

- **Ask what you have invested.** Money, time, public statements or self-image can raise confidence without adding evidence. *(Chapter 1)*
- **Name one thing that would change your mind.** If you cannot, your belief may be functioning as a badge. *(Chapters 1 and 2)*

When you evaluate evidence

- **Use the same standard for both sides.** Swap "Can I believe this?" and "Must I believe this?" *(Chapter 3)*
- **Consider the opposite.** Would you judge this study the same way if it had found the reverse? *(Chapter 8)*
- **Search neutrally.** Ask "Is it true that…?" rather than "Why is it true that…?" *(Chapter 3)*

When you rely on memory

- **Trust records over recollection.** A note written at the time beats a vivid memory recalled later. *(Chapter 4)*
- **Be suspicious of flattering memories.** If your version makes you look unusually wise, check it. *(Chapter 4)*

When you feel clever

- **Argue the other side well.** Write your opponent's best case, not a caricature. *(Chapter 5)*
- **Pretend your argument is a stranger's.** Evaluate it as critically as you would theirs. *(Chapter 5)*

- **Keep score.** Make specific, dated predictions with probabilities and check them. (*Chapters 5 and 8*)

When you are in a group

- **Look for pluralistic ignorance.** Others may privately share your doubts. (*Chapter 6*)
- **Check whether you would hold this view if your group did not.** (*Chapter 6*)

Before you forward

- **Pause, rate your confidence and name the emotion.** (*Chapter 7*)
- **Check the original source.** (*Chapter 7*)
- **Ask who could be harmed if it is false.** (*Chapter 7*)

When talking to someone who disagrees

- **Don't humiliate. Ask, don't tell. Lead with shared values.** (*Chapter 8*)
- **Replace a false explanation with a true one**, rather than simply denying it. (*Chapter 7*)
- **Be patient.** Plant a seed; do not demand a confession. (*Chapter 8*)

Every month

- **Update your log.** Record what you have changed your mind about, and celebrate it. (*Chapter 8*)

Notes and Sources

These notes list the main research and reporting behind each chapter, for readers who want to go to the original sources. Composite characters (identified in the text) are illustrative and are not based on specific individuals.

Introduction

- Kunda, Z. (1990). The case for motivated reasoning. *Psychological Bulletin*, 108(3), 480–498.
- Haidt, J. (2012). *The Righteous Mind: Why Good People Are Divided by Politics and Religion*. Pantheon.
- Kahneman, D. (2011). *Thinking, Fast and Slow*. Farrar, Straus and Giroux.
- Open Science Collaboration (2015). Estimating the reproducibility of psychological science. *Science*, 349(6251).

Chapter 1 — The Need to Be Right

- Festinger, L., Riecken, H. W., & Schachter, S. (1956). *When Prophecy Fails*. University of Minnesota Press.
- Festinger, L., & Carlsmith, J. M. (1959). Cognitive consequences of forced compliance. *Journal of Abnormal and Social Psychology*, 58(2), 203–210.
- Aronson, E., & Mills, J. (1959). The effect of severity of initiation on liking for a group. *Journal of Abnormal and Social Psychology*, 59(2), 177–181.
- Knox, R. E., & Inkster, J. A. (1968). Postdecision dissonance at post time. *Journal of Personality and Social Psychology*, 8(4), 319–323.
- Deutsch, M., & Gerard, H. B. (1955). A study of normative and informational social influences upon individual judgment. *Journal of Abnormal and Social Psychology*, 51(3), 629–636.
- Tavris, C., & Aronson, E. (2007/2015). *Mistakes Were Made (But Not by Me)*. Harcourt.
- Burton, R. A. (2008). *On Being Certain*. St. Martin's Press.
- Bem, D. J. (1967). Self-perception: An alternative interpretation of cognitive dissonance phenomena. *Psychological Review*, 74(3), 183–200.

- Satyam: reporting on B. Ramalinga Raju's letter of 7 January 2009 and its later repudiation, *The News Minute* and *The Quint* (March 2015); *Compliance Week*; Vivek Kaul's columns on the Satyam case.

Chapter 2 — How Beliefs Protect Identity

- Kahan, D. M., et al. (2012). The polarizing impact of science literacy and numeracy on perceived climate change risks. *Nature Climate Change*, 2, 732–735.
- Kahan, D. M. (2013). Ideology, motivated reasoning, and cognitive reflection. *Judgment and Decision Making*, 8(4), 407–424.
- Cohen, G. L., Aronson, J., & Steele, C. M. (2000). When beliefs yield to evidence: Reducing biased evaluation by affirming the self. *Personality and Social Psychology Bulletin*, 26(9), 1151–1164.
- Jones, E. E., & Harris, V. A. (1967). The attribution of attitudes. *Journal of Experimental Social Psychology*, 3(1), 1–24.
- Miller, J. G. (1984). Culture and the development of everyday social explanation. *Journal of Personality and Social Psychology*, 46(5), 961–978.
- Mezulis, A. H., Abramson, L. Y., Hyde, J. S., & Hankin, B. L. (2004). Is there a universal positivity bias in attributions? *Psychological Bulletin*, 130(5), 711–747.
- McAdams, D. P. (2006). *The Redemptive Self.* Oxford University Press.
- Semmelweis, I. (1861/1983). *The Etiology, Concept, and Prophylaxis of Childbed Fever* (trans. K. C. Carter). University of Wisconsin Press.

Chapter 3 — The Evidence We Choose to See

- Wason, P. C. (1960). On the failure to eliminate hypotheses in a conceptual task. *Quarterly Journal of Experimental Psychology*, 12(3), 129–140.
- Klayman, J., & Ha, Y.-W. (1987). Confirmation, disconfirmation, and information in hypothesis testing. *Psychological Review*, 94(2), 211–228.
- Hastorf, A. H., & Cantril, H. (1954). They saw a game: A case study. *Journal of Abnormal and Social Psychology*, 49(1), 129–134.
- Gilovich, T. (1991). *How We Know What Isn't So.* Free Press.
- Ditto, P. H., & Lopez, D. F. (1992). Motivated skepticism. *Journal of Personality and Social Psychology*, 63(4), 568–584.
- Lord, C. G., Ross, L., & Lepper, M. R. (1979). Biased assimilation and attitude polarization. *Journal of Personality and Social Psychology*, 37(11), 2098–2109.
- Kuhn, D., & Lao, J. (1996). Effects of evidence on attitudes: Is polarization the norm? *Psychological Science*, 7(2), 115–120.
- Hart, W., et al. (2009). Feeling validated versus being correct: A meta-analysis of selective exposure to information. *Psychological Bulletin*, 135(4), 555–588.
- Mercier, H., & Sperber, D. (2011). Why do humans reason? *Behavioral and Brain Sciences*, 34(2), 57–74.

- Ganesha milk "miracle", 21 September 1995: contemporaneous coverage in *The Times of India*, *The Guardian*, *The Independent* and others; summary accounts in *The Wire* (2020) and *Mental Floss*; rationalist explanations by the Maharashtra Andhashraddha Nirmulan Samiti.

Chapter 4 — Why Memory Is Not a Recording

- Thompson-Cannino, J., Cotton, R., & Torneo, E. (2009). *Picking Cotton.* St. Martin's Press.
- Bartlett, F. C. (1932). *Remembering.* Cambridge University Press.
- Loftus, E. F., & Palmer, J. C. (1974). Reconstruction of automobile destruction. *Journal of Verbal Learning and Verbal Behavior*, 13(5), 585–589.
- Loftus, E. F., & Pickrell, J. E. (1995). The formation of false memories. *Psychiatric Annals*, 25(12), 720–725.
- Scoboria, A., et al. (2017). A mega-analysis of memory reports from eight peer-reviewed false memory implantation studies. *Memory*, 25(2), 146–163.
- Neisser, U., & Harsch, N. (1992). Phantom flashbulbs: False recollections of hearing the news about Challenger. In E. Winograd & U. Neisser (Eds.), *Affect and Accuracy in Recall.* Cambridge University Press.
- Talarico, J. M., & Rubin, D. C. (2003). Confidence, not consistency, characterizes flashbulb memories. *Psychological Science*, 14(5), 455–461.
- Bahrick, H. P., Hall, L. K., & Berger, S. A. (1996). Accuracy and distortion in memory for high school grades. *Psychological Science*, 7(5), 265–271.
- Fischhoff, B. (1975). Hindsight ≠ foresight. *Journal of Experimental Psychology: Human Perception and Performance*, 1(3), 288–299.
- Wells, G. L., & Bradfield, A. L. (1998). "Good, you identified the suspect": Feedback to eyewitnesses distorts their reports. *Journal of Applied Psychology*, 83(3), 360–376.
- Wixted, J. T., & Wells, G. L. (2017). The relationship between eyewitness confidence and identification accuracy. *Psychological Science in the Public Interest*, 18(1), 10–65.
- Innocence Project, eyewitness misidentification resources (innocenceproject.org).

Chapter 5 — When Intelligence Becomes a Defence Mechanism

- Doyle, A. C. (1922). *The Coming of the Fairies.* Hodder & Stoughton. On the 1983 confessions, see contemporary reporting and Cooper, J. (1990), *The Case of the Cottingley Fairies.*
- Stanovich, K. E., West, R. F., & Toplak, M. E. (2013). Myside bias, rational thinking, and intelligence. *Current Directions in Psychological Science*, 22(4), 259–264.

- Taber, C. S., & Lodge, M. (2006). Motivated skepticism in the evaluation of political beliefs. *American Journal of Political Science*, 50(3), 755–769.
- Kahan, D. M., Peters, E., Dawson, E. C., & Slovic, P. (2017). Motivated numeracy and enlightened self-government. *Behavioural Public Policy*, 1(1), 54–86.
- Persson, E., et al. (2021). A preregistered replication of motivated numeracy. *Cognition*, 214, 104768.
- Pronin, E., Lin, D. Y., & Ross, L. (2002). The bias blind spot. *Personality and Social Psychology Bulletin*, 28(3), 369–381.
- West, R. F., Meserve, R. J., & Stanovich, K. E. (2012). Cognitive sophistication does not attenuate the bias blind spot. *Journal of Personality and Social Psychology*, 103(3), 506–519.
- Tetlock, P. E. (2005). *Expert Political Judgment.* Princeton University Press.
- Tetlock, P. E., & Gardner, D. (2015). *Superforecasting.* Crown.
- Trouche, E., Johansson, P., Hall, L., & Mercier, H. (2016). The selective laziness of reasoning. *Cognitive Science*, 40(8), 2122–2136.

Chapter 6 — Groups, Politics and Shared Illusions

- Mahim Creek, 18–19 August 2006: reporting by Reuters, *The Tribune*, UNI and others; analysis by Praful Bidwai.
- Asch, S. E. (1956). Studies of independence and conformity. *Psychological Monographs*, 70(9).
- Bond, R., & Smith, P. B. (1996). Culture and conformity. *Psychological Bulletin*, 119(1), 111–137.
- Prentice, D. A., & Miller, D. T. (1993). Pluralistic ignorance and alcohol use on campus. *Journal of Personality and Social Psychology*, 64(2), 243–256.
- Schkade, D., Sunstein, C. R., & Hastie, R. (2007). What happened on deliberation day? *California Law Review*, 95(3), 915–940.
- Tajfel, H., Billig, M. G., Bundy, R. P., & Flament, C. (1971). Social categorization and intergroup behaviour. *European Journal of Social Psychology*, 1(2), 149–178.
- Cohen, G. L. (2003). Party over policy. *Journal of Personality and Social Psychology*, 85(5), 808–822.
- Iyengar, S., & Westwood, S. J. (2015). Fear and loathing across party lines. *American Journal of Political Science*, 59(3), 690–707.
- Janis, I. L. (1972). *Victims of Groupthink.* Houghton Mifflin.
- Guess, A. M. (2021). (Almost) everything in moderation: New evidence on Americans' online media diets. *American Journal of Political Science*, 65(4), 1007–1022.
- Le Texier, T. (2019). Debunking the Stanford Prison Experiment. *American Psychologist*, 74(7), 823–839.

- Monkey Man of Delhi (2001): Verma, S. K., & Srivastava, D. K. (2003). A study on mass hysteria (monkey men?) victims in East Delhi. *Indian Journal of Medical Sciences*, 57(8), 355–360; reporting by the BBC, *The Guardian* and Rediff (May 2001).

Chapter 7 — How False Stories Spread

- Dhule/Rainpada lynching, 1 July 2018, and related incidents: reporting by BOOM, *The Quint*, AFP and *Governance Now*.
- Allport, G. W., & Postman, L. (1947). *The Psychology of Rumor.* Henry Holt.
- Vosoughi, S., Roy, D., & Aral, S. (2018). The spread of true and false news online. *Science*, 359(6380), 1146–1151.
- Brady, W. J., et al. (2017). Emotion shapes the diffusion of moralized content in social networks. *PNAS*, 114(28), 7313–7318.
- Hasher, L., Goldstein, D., & Toppino, T. (1977). Frequency and the conference of referential validity. *Journal of Verbal Learning and Verbal Behavior*, 16(1), 107–112.
- Fazio, L. K., Brashier, N. M., Payne, B. K., & Marsh, E. J. (2015). Knowledge does not protect against illusory truth. *Journal of Experimental Psychology: General*, 144(5), 993–1002.
- Demonetisation rumours (November 2016): reporting by *The News Minute*, *YourStory*, *Moneylife*/IANS and *The Wire* Science.
- Johnson, H. M., & Seifert, C. M. (1994). Sources of the continued influence effect. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 20(6), 1420–1436.
- Lewandowsky, S., Ecker, U. K. H., et al. (2012). Misinformation and its correction. *Psychological Science in the Public Interest*, 13(3), 106–131.
- Nyhan, B., & Reifler, J. (2010). When corrections fail. *Political Behavior*, 32(2), 303–330.
- Wood, T., & Porter, E. (2019). The elusive backfire effect. *Political Behavior*, 41(1), 135–163.
- Pennycook, G., & Rand, D. G. (2019). Lazy, not biased. *Cognition*, 188, 39–50.
- Pennycook, G., et al. (2021). Shifting attention to accuracy can reduce misinformation online. *Nature*, 592, 590–595.
- Arechar, A. A., et al. (2023). Understanding and combatting misinformation across 16 countries on six continents. *Nature Human Behaviour*, 7, 1502–1513.
- WhatsApp, "More changes to forwarding" (blog post, 19 July 2018; updated 21 January 2019).

Chapter 8 — The Difficult Art of Changing Your Mind

- Broockman, D., Kalla, J., & Aronow, P. (2015). Irregularities in LaCour (2014). Working paper.

- Broockman, D., & Kalla, J. (2016). Durably reducing transphobia: A field experiment on door-to-door canvassing. *Science*, 352(6282), 220–224.
- Kalla, J. L., & Broockman, D. E. (2020). Reducing exclusionary attitudes through interpersonal conversation. *American Political Science Review*, 114(2), 410–425.
- Galef, J. (2021). *The Scout Mindset*. Portfolio.
- Leary, M. R., et al. (2017). Cognitive and interpersonal features of intellectual humility. *Personality and Social Psychology Bulletin*, 43(6), 793–813.
- Lord, C. G., Lepper, M. R., & Preston, E. (1984). Considering the opposite. *Journal of Personality and Social Psychology*, 47(6), 1231–1243.
- Darwin, C. (1887/1958). *The Autobiography of Charles Darwin* (ed. N. Barlow). Collins.
- McGuire, W. J. (1964). Inducing resistance to persuasion. *Advances in Experimental Social Psychology*, 1, 191–229.
- Roozenbeek, J., van der Linden, S., Goldberg, B., Rathje, S., & Lewandowsky, S. (2022). Psychological inoculation improves resilience against misinformation on social media. *Science Advances*, 8(34).
- Jigsaw prebunking campaign in India (Bengali, Hindi, Marathi): Reuters, December 2022.
- Kannur *Satyameva Jayate* initiative: *The News Minute* (June 2018), *The Logical Indian* (August 2018), BBC coverage reported by *Beebom* (2018).

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